ML Architecture

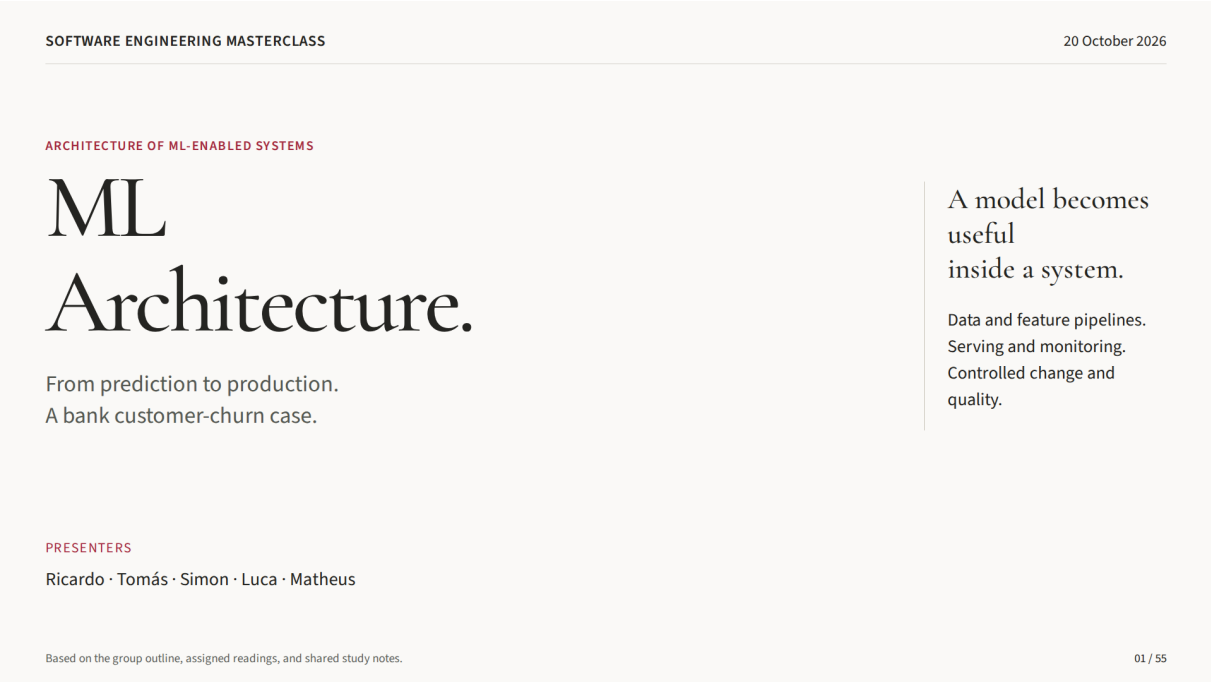
Speaker Notes

A bank customer-churn case — 20 October 2026

Each page shows one slide exactly as presenters will see it, the words to speak, and short delivery cues. The script is a speaking aid, not a transcript to read verbatim — adapt it to your own voice. Animated builds are marked; press the arrow key the stated number of times before moving to the next slide.

01 / 55 ML Architecture

Block 1 / Foundations • All presenters



TO SAY

- From a trained model to a maintainable, observable, governed system.
- **Software engineering** for systems that depend on learned behavior.
- **One shared case:** bank customer-churn prediction.
- **Two blocks:** model-to-system foundations, then architecture and quality.
- **Five presenters:** Ricardo, Tomás, Simon, Luca, and Matheus.
- *At our bank:* Bank churn supplies the shared case. Examples, records, schedules, and incidents are illustrative.
- *A useful model needs an engineered system around it.*

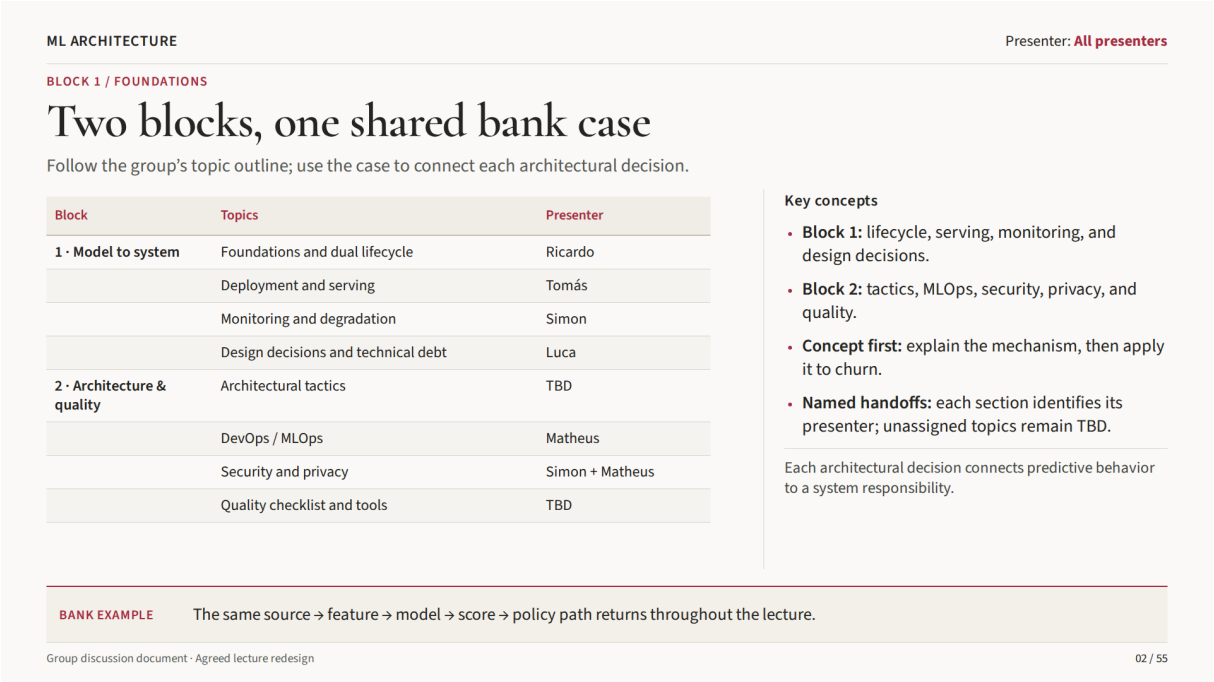
CUES

- New block: **Block 1 / Foundations**.
- Presenter: **All presenters**.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

Reading anchor: Bass, Ch. 14 · Kalinowski, Ch. 13–14

02 / 55 Two blocks, one shared bank case

Block 1 / Foundations • All presenters



ML ARCHITECTURE

Presenter: All presenters

BLOCK 1 / FOUNDATIONS

Two blocks, one shared bank case

Follow the group's topic outline; use the case to connect each architectural decision.

Block	Topics	Presenter
1 · Model to system	Foundations and dual lifecycle	Ricardo
	Deployment and serving	Tomás
	Monitoring and degradation	Simon
	Design decisions and technical debt	Luca
2 · Architecture & quality	Architectural tactics	TBD
	DevOps / MLOps	Matheus
	Security and privacy	Simon + Matheus
	Quality checklist and tools	TBD

Key concepts

- **Block 1:** lifecycle, serving, monitoring, and design decisions.
- **Block 2:** tactics, MLOps, security, privacy, and quality.
- **Concept first:** explain the mechanism, then apply it to churn.
- **Named handoffs:** each section identifies its presenter; unassigned topics remain TBD.

Each architectural decision connects predictive behavior to a system responsibility.

BANK EXAMPLE The same source → feature → model → score → policy path returns throughout the lecture.

Group discussion document · Agreed lecture redesign 02 / 55

TO SAY

- Follow the group's topic outline; use the case to connect each architectural decision.
- **Block 1:** lifecycle, serving, monitoring, and design decisions.
- **Block 2:** tactics, MLOps, security, privacy, and quality.
- **Concept first:** explain the mechanism, then apply it to churn.
- **Named handoffs:** each section identifies its presenter; unassigned topics remain TBD.
- *At our bank:* The same source → feature → model → score → policy path returns throughout the lecture.
- *Each architectural decision connects predictive behavior to a system responsibility.*

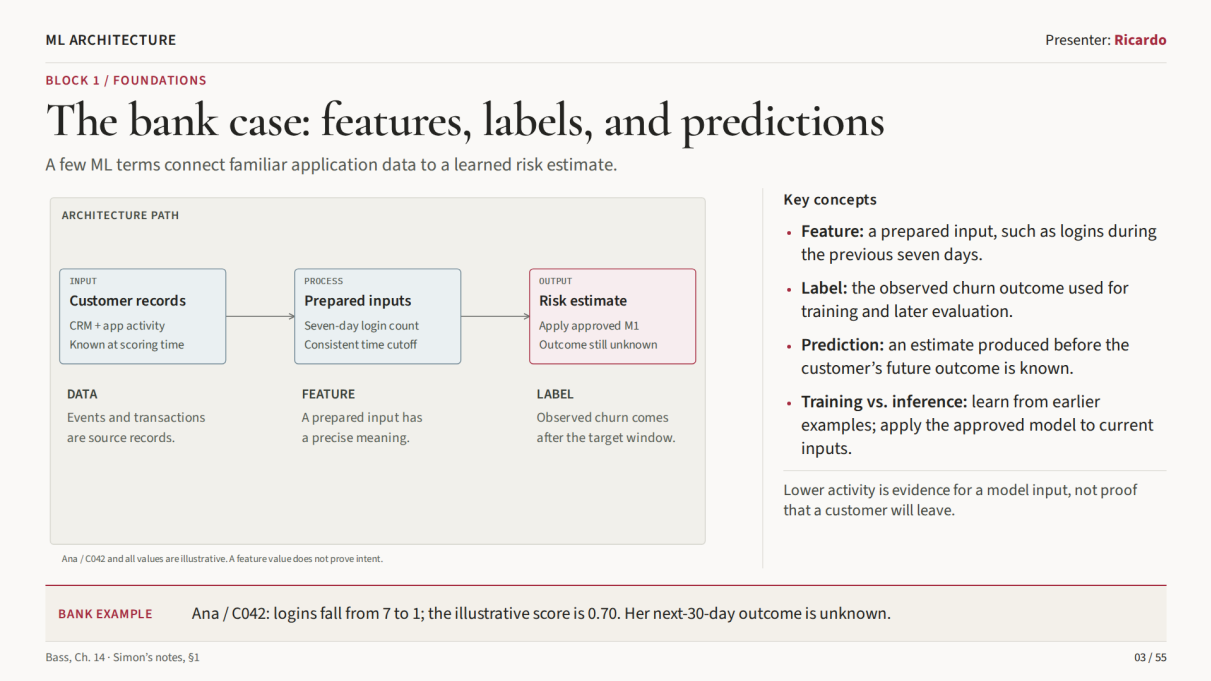
CUES

- Presenter: **All presenters**.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

Reading anchor: Group discussion document · Agreed lecture redesign

03 / 55 The bank case: features, labels, and predictions

Block 1 / Foundations • Ricardo



TO SAY

- A few ML terms connect familiar application data to a learned risk estimate.
- **Feature:** a prepared input, such as logins during the previous seven days.
- **Label:** the observed churn outcome used for training and later evaluation.
- **Prediction:** an estimate produced before the customer's future outcome is known.
- **Training vs. inference:** learn from earlier examples; apply the approved model to current inputs.
- *At our bank:* Ana / C042: logins fall from 7 to 1; the illustrative score is 0.70. Her next-30-day outcome is unknown.
- *Lower activity is evidence for a model input, not proof that a customer will leave.*

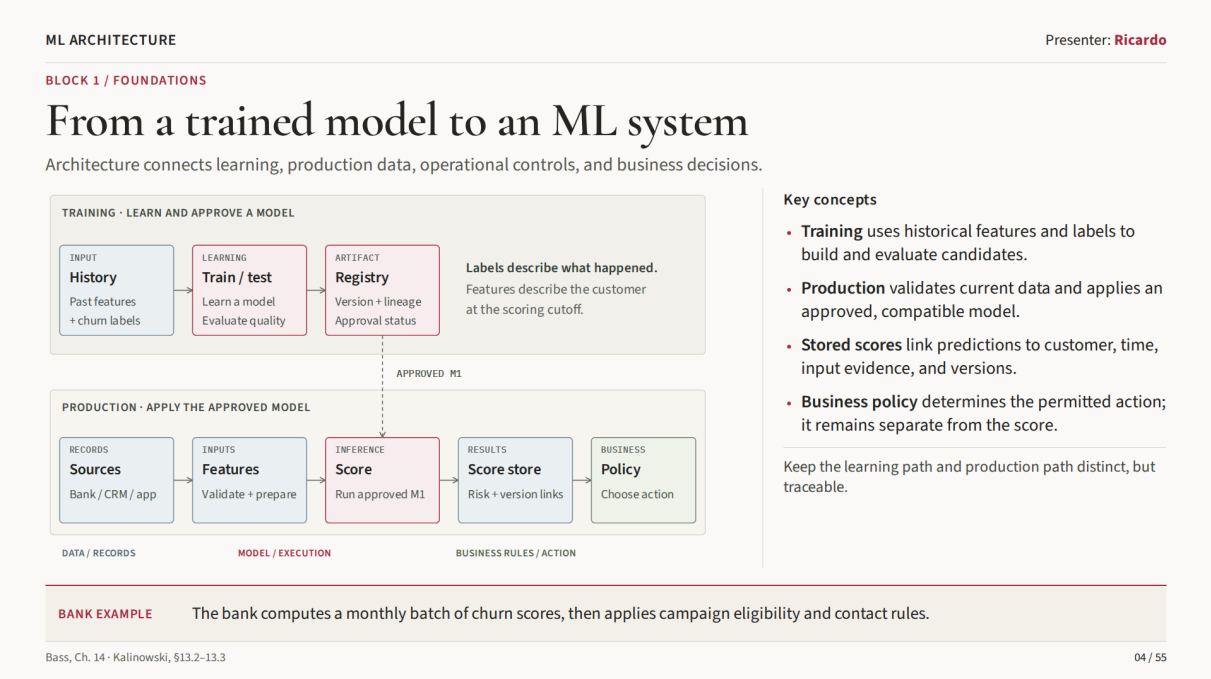
CUES

- Presenter: **Ricardo**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

Reading anchor: Bass, Ch. 14 · Simon's notes, §1

04 / 55 From a trained model to an ML system

Block 1 / Foundations • Ricardo



TO SAY

- Architecture connects learning, production data, operational controls, and business decisions.
- **Training** uses historical features and labels to build and evaluate candidates.
- **Production** validates current data and applies an approved, compatible model.
- **Stored scores** link predictions to customer, time, input evidence, and versions.
- **Business policy** determines the permitted action; it remains separate from the score.
- *At our bank:* The bank computes a monthly batch of churn scores, then applies campaign eligibility and contact rules.
- *Keep the learning path and production path distinct, but traceable.*

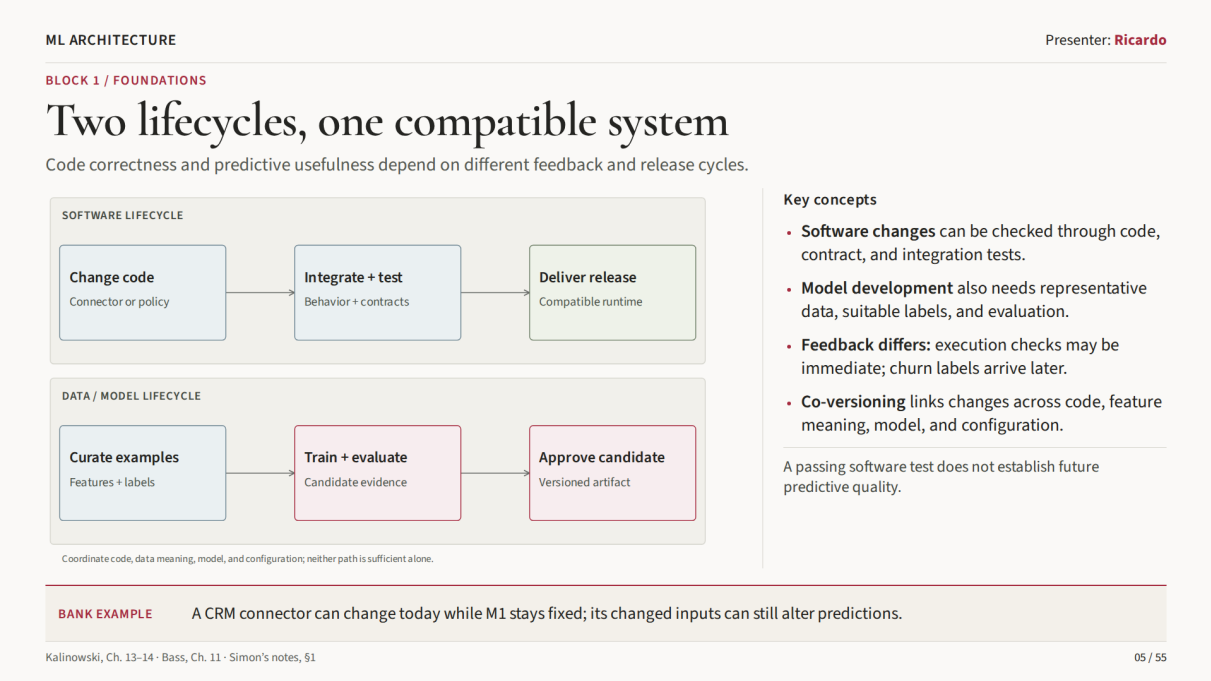
CUES

- Presenter: **Ricardo**.
- Animated build: click 4 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

Reading anchor: Bass, Ch. 14 · Kalinowski, §13.2–13.3

05 / 55 Two lifecycles, one compatible system

Block 1 / Foundations • Ricardo



TO SAY

- Code correctness and predictive usefulness depend on different feedback and release cycles.
- **Software changes** can be checked through code, contract, and integration tests.
- **Model development** also needs representative data, suitable labels, and evaluation.
- **Feedback differs:** execution checks may be immediate; churn labels arrive later.
- **Co-versioning** links changes across code, feature meaning, model, and configuration.
- *At our bank:* A CRM connector can change today while M1 stays fixed; its changed inputs can still alter predictions.
- *A passing software test does not establish future predictive quality.*

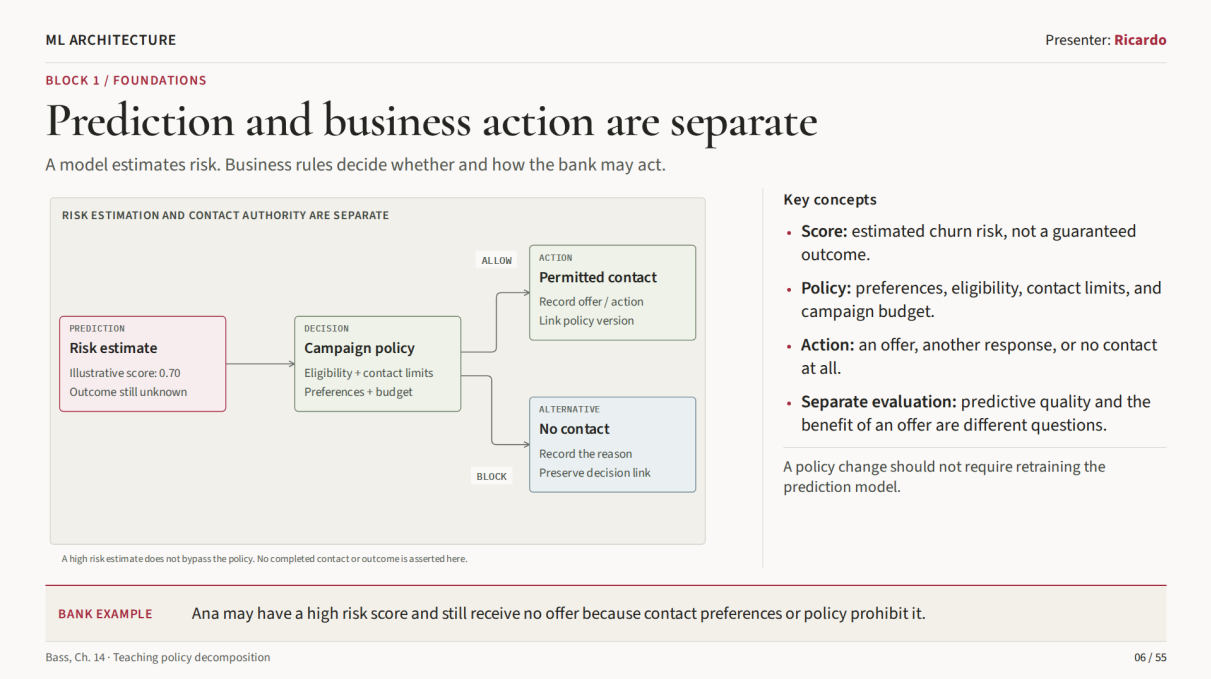
CUES

- Presenter: **Ricardo**.
- Animated build: click 3 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

Reading anchor: Kalinowski, Ch. 13–14 · Bass, Ch. 11 · Simon's notes, §1

06 / 55 Prediction and business action are separate

Block 1 / Foundations • Ricardo



TO SAY

- A model estimates risk. Business rules decide whether and how the bank may act.
- **Score:** estimated churn risk, not a guaranteed outcome.
- **Policy:** preferences, eligibility, contact limits, and campaign budget.
- **Action:** an offer, another response, or no contact at all.
- **Separate evaluation:** predictive quality and the benefit of an offer are different questions.
- *At our bank:* Ana may have a high risk score and still receive no offer because contact preferences or policy prohibit it.
- *A policy change should not require retraining the prediction model.*

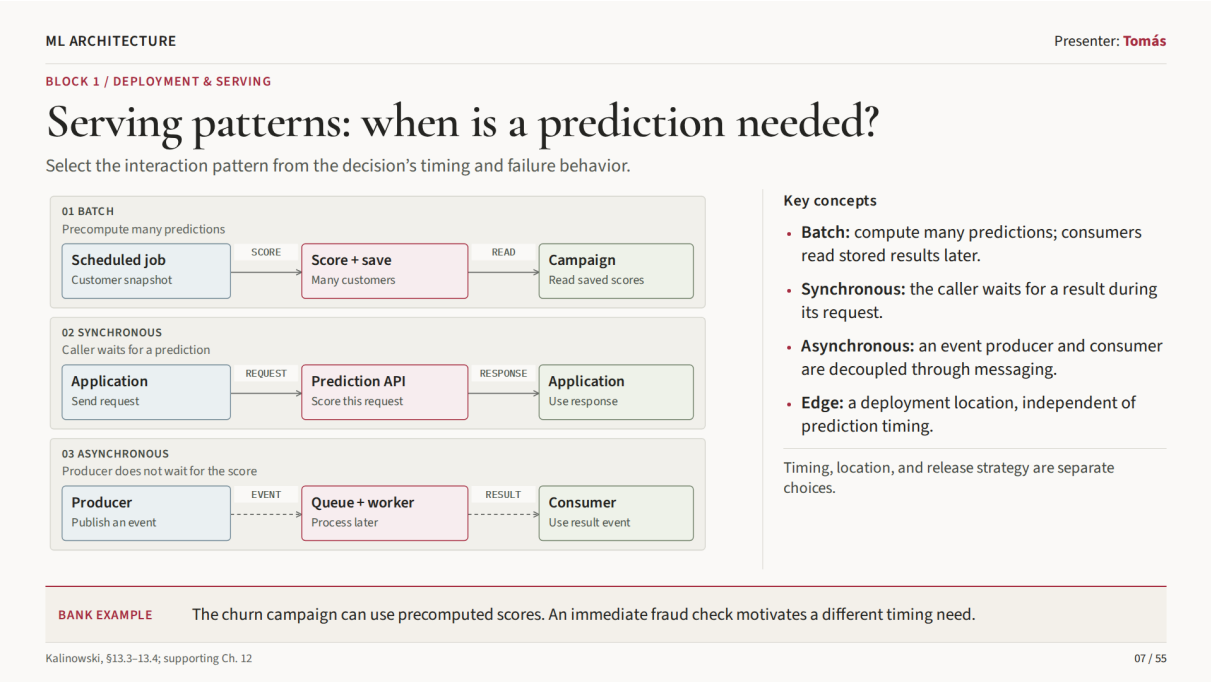
CUES

- Presenter: **Ricardo**.
- Animated build: click 3 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

Reading anchor: Bass, Ch. 14 · Teaching policy decomposition

07 / 55 Serving patterns: when is a prediction needed?

Block 1 / Deployment & serving • Tomás



TO SAY

- Select the interaction pattern from the decision's timing and failure behavior.
- **Batch:** compute many predictions; consumers read stored results later.
- **Synchronous:** the caller waits for a result during its request.
- **Asynchronous:** an event producer and consumer are decoupled through messaging.
- **Edge:** a deployment location, independent of prediction timing.
- *At our bank:* The churn campaign can use precomputed scores. An immediate fraud check motivates a different timing need.
- *Timing, location, and release strategy are separate choices.*

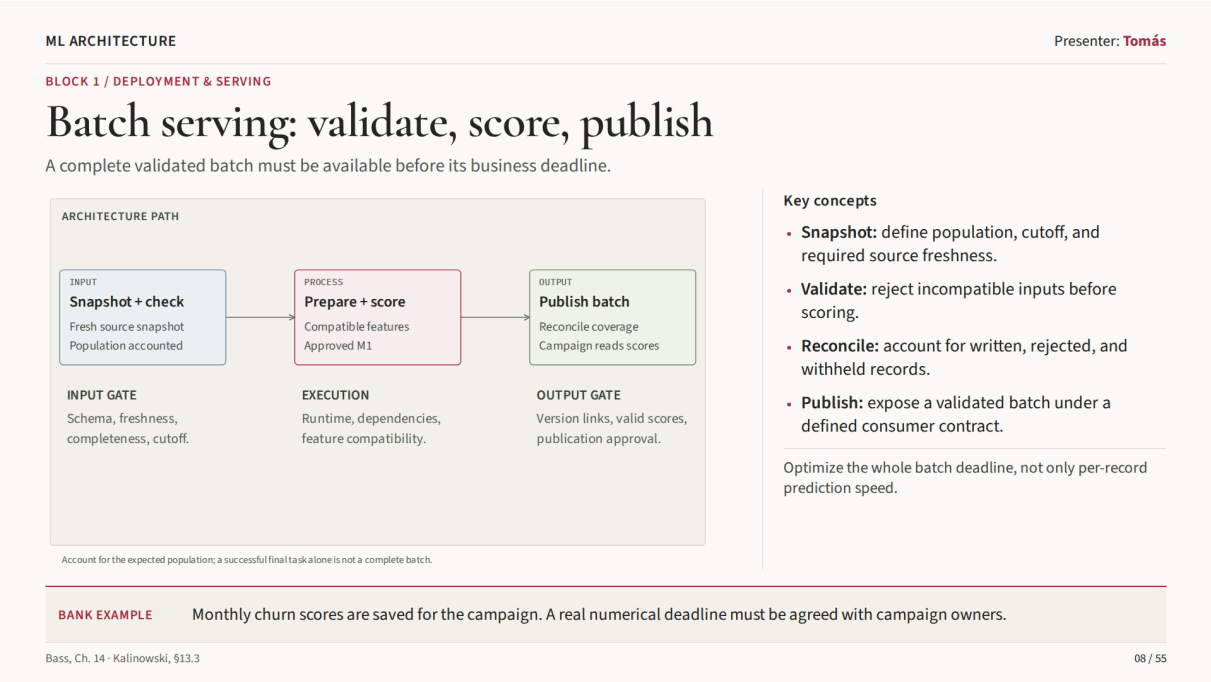
CUES

- New block: **Block 1 / Deployment & serving**.
- Presenter: **Tomás**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

Reading anchor: Kalinowski, §13.3–13.4; supporting Ch. 12

08 / 55 Batch serving: validate, score, publish

Block 1 / Deployment & serving • Tomás



TO SAY

- A complete validated batch must be available before its business deadline.
- **Snapshot:** define population, cutoff, and required source freshness.
- **Validate:** reject incompatible inputs before scoring.
- **Reconcile:** account for written, rejected, and withheld records.
- **Publish:** expose a validated batch under a defined consumer contract.
- *At our bank:* Monthly churn scores are saved for the campaign. A real numerical deadline must be agreed with campaign owners.
- *Optimize the whole batch deadline, not only per-record prediction speed.*

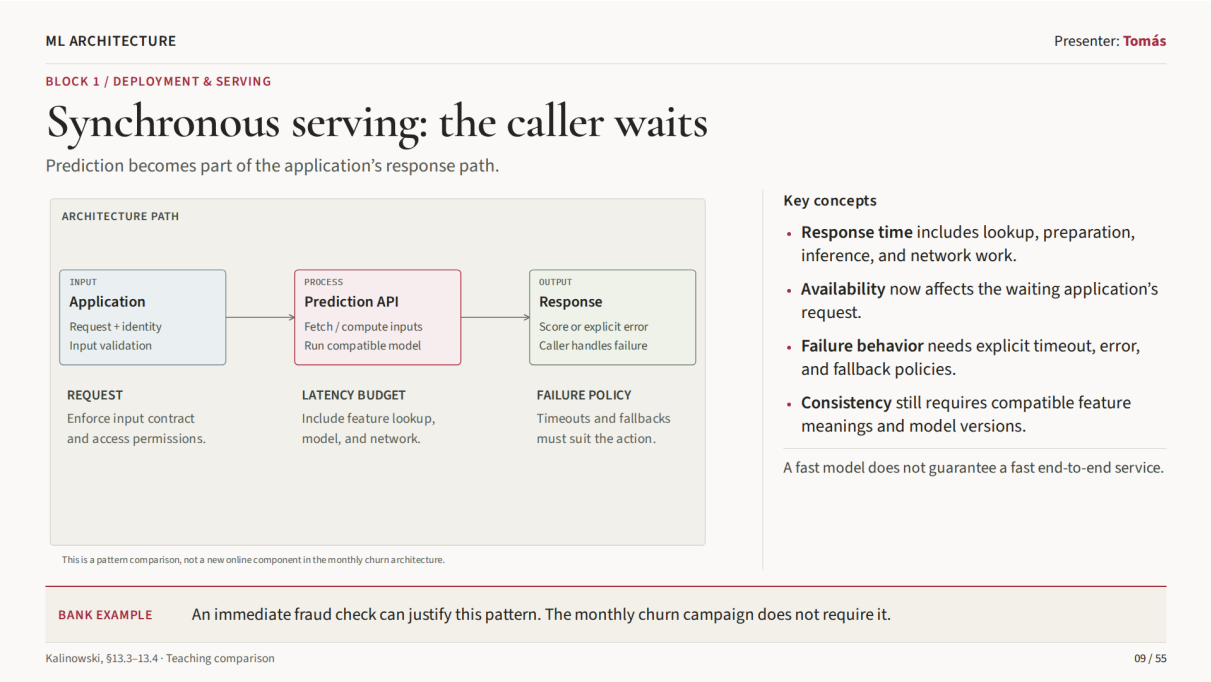
CUES

- Presenter: **Tomás**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

Reading anchor: Bass, Ch. 14 · Kalinowski, §13.3

09 / 55 Synchronous serving: the caller waits

Block 1 / Deployment & serving • Tomás



TO SAY

- Prediction becomes part of the application's response path.
- **Response time** includes lookup, preparation, inference, and network work.
- **Availability** now affects the waiting application's request.
- **Failure behavior** needs explicit timeout, error, and fallback policies.
- **Consistency** still requires compatible feature meanings and model versions.
- *At our bank:* An immediate fraud check can justify this pattern. The monthly churn campaign does not require it.
- *A fast model does not guarantee a fast end-to-end service.*

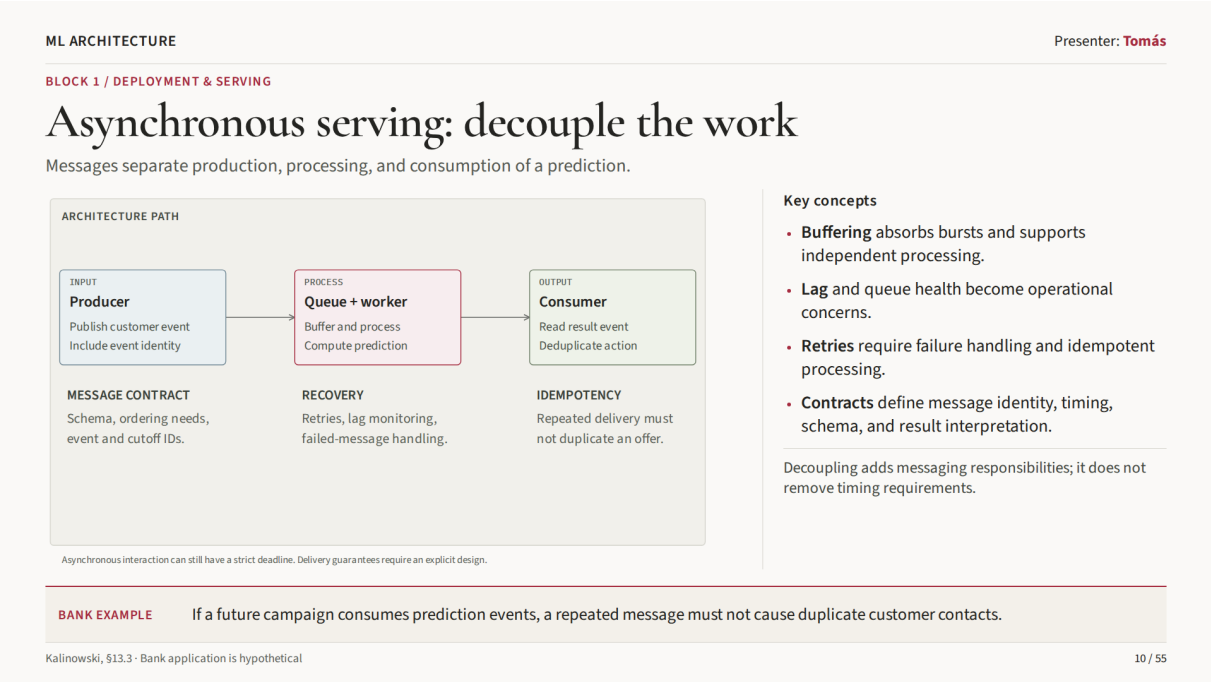
CUES

- Presenter: **Tomás**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

Reading anchor: Kalinowski, §13.3–13.4 · Teaching comparison

10 / 55 Asynchronous serving: decouple the work

Block 1 / Deployment & serving • Tomás



TO SAY

- Messages separate production, processing, and consumption of a prediction.
- **Buffering** absorbs bursts and supports independent processing.
- **Lag** and queue health become operational concerns.
- **Retries** require failure handling and idempotent processing.
- **Contracts** define message identity, timing, schema, and result interpretation.
- *At our bank:* If a future campaign consumes prediction events, a repeated message must not cause duplicate customer contacts.
- *Decoupling adds messaging responsibilities; it does not remove timing requirements.*

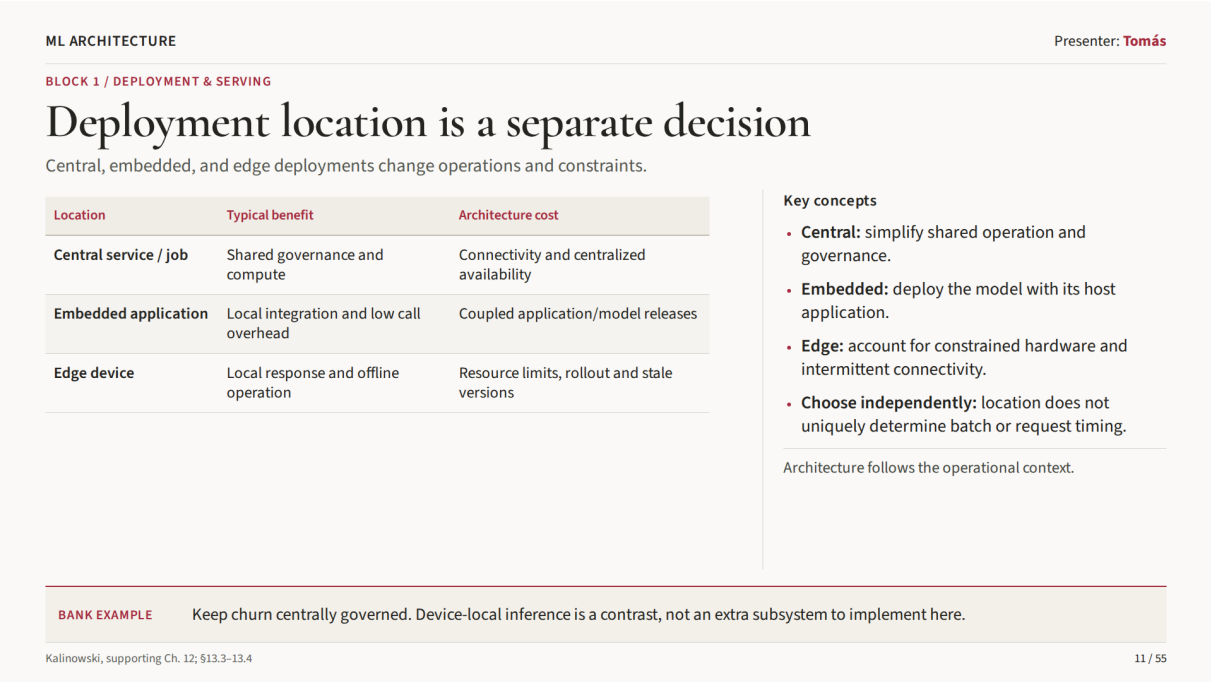
CUES

- Presenter: **Tomás**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

Reading anchor: Kalinowski, §13.3 · Bank application is hypothetical

11 / 55 Deployment location is a separate decision

Block 1 / Deployment & serving • Tomás



TO SAY

- Central, embedded, and edge deployments change operations and constraints.
- **Central:** simplify shared operation and governance.
- **Embedded:** deploy the model with its host application.
- **Edge:** account for constrained hardware and intermittent connectivity.
- **Choose independently:** location does not uniquely determine batch or request timing.
- *At our bank:* Keep churn centrally governed. Device-local inference is a contrast, not an extra subsystem to implement here.
- *Architecture follows the operational context.*

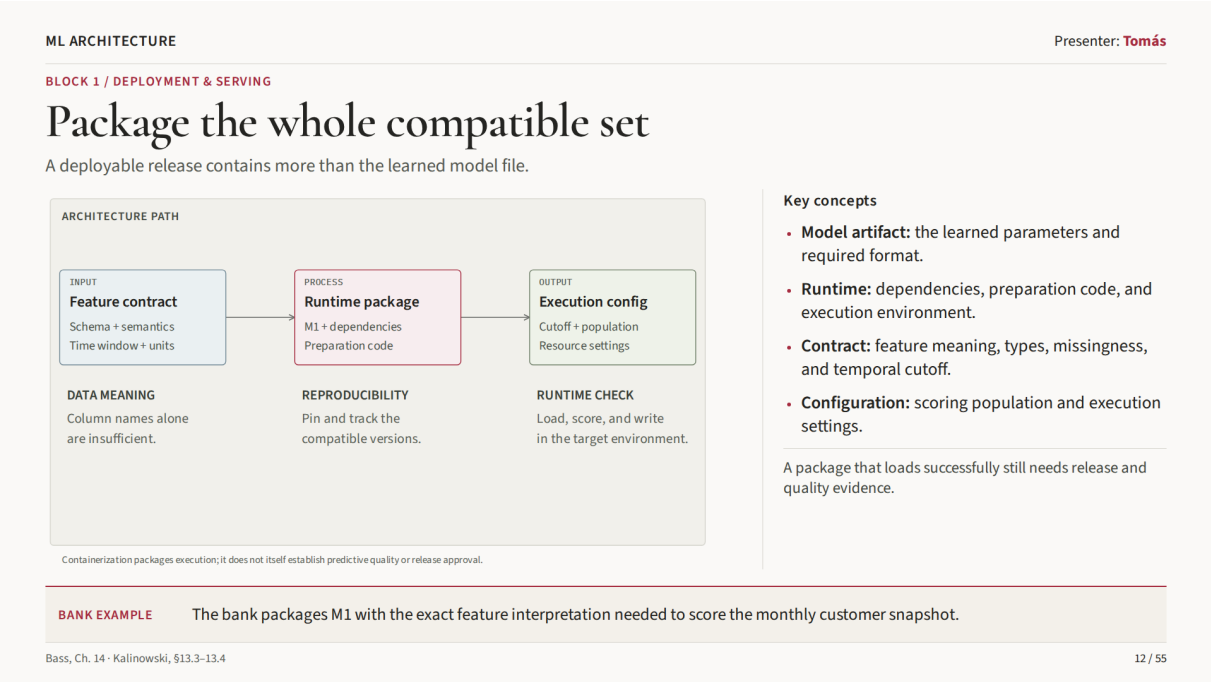
CUES

- Presenter: **Tomás**.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

Reading anchor: Kalinowski, supporting Ch. 12; §13.3–13.4

12 / 55 Package the whole compatible set

Block 1 / Deployment & serving • Tomás



TO SAY

- A deployable release contains more than the learned model file.
- **Model artifact:** the learned parameters and required format.
- **Runtime:** dependencies, preparation code, and execution environment.
- **Contract:** feature meaning, types, missingness, and temporal cutoff.
- **Configuration:** scoring population and execution settings.
- *At our bank:* The bank packages M1 with the exact feature interpretation needed to score the monthly customer snapshot.
- *A package that loads successfully still needs release and quality evidence.*

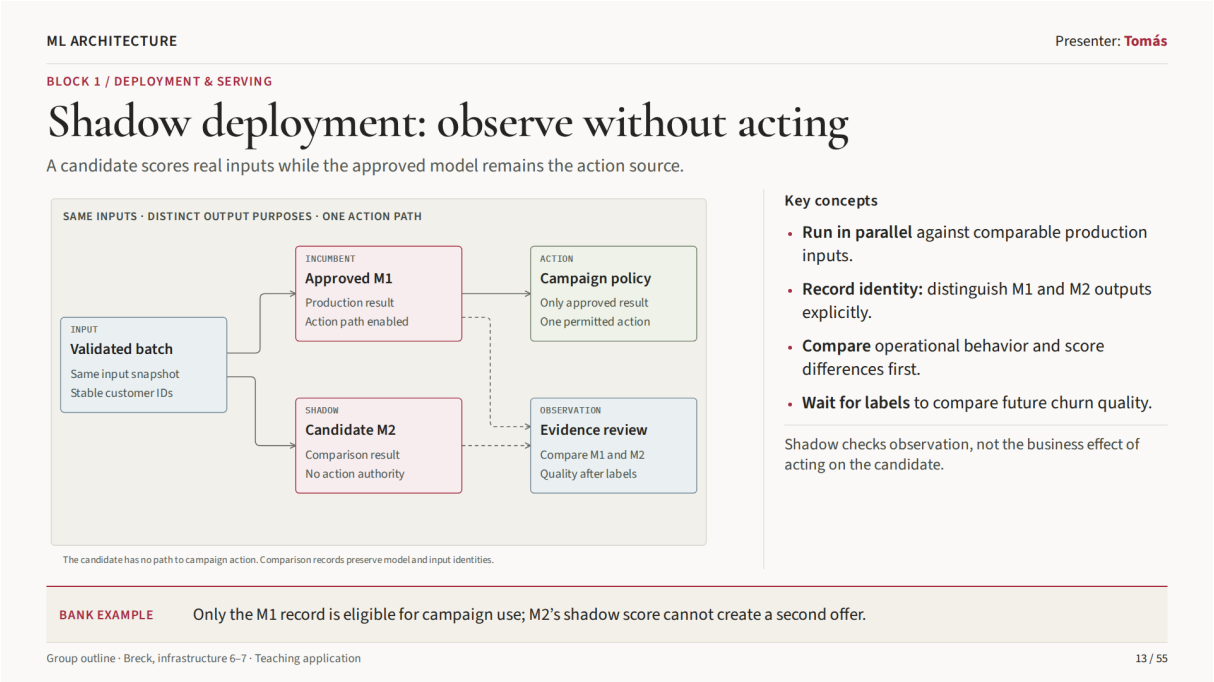
CUES

- Presenter: **Tomás**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
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Reading anchor: Bass, Ch. 14 · Kalinowski, §13.3–13.4

13 / 55 Shadow deployment: observe without acting

Block 1 / Deployment & serving • Tomás



TO SAY

- A candidate scores real inputs while the approved model remains the action source.
- **Run in parallel** against comparable production inputs.
- **Record identity:** distinguish M1 and M2 outputs explicitly.
- **Compare** operational behavior and score differences first.
- **Wait for labels** to compare future churn quality.
- *At our bank:* Only the M1 record is eligible for campaign use; M2's shadow score cannot create a second offer.
- *Shadow checks observation, not the business effect of acting on the candidate.*

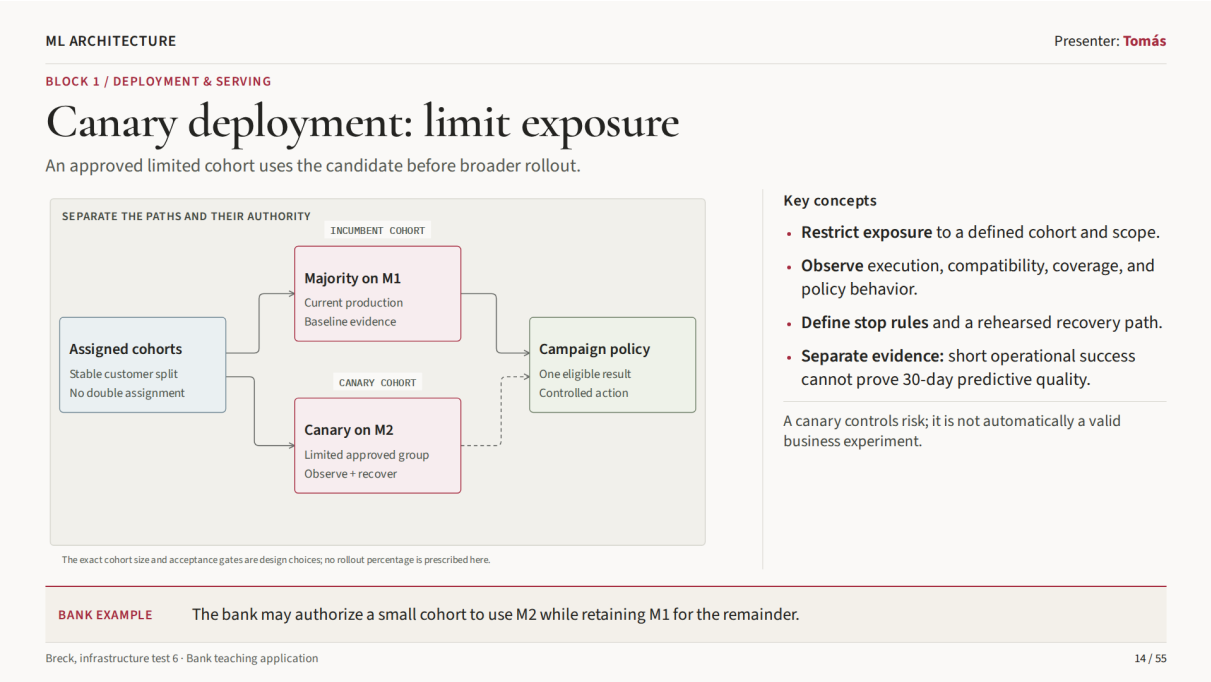
CUES

- Presenter: **Tomás**.
- Animated build: click 4 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

Reading anchor: Group outline · Breck, infrastructure 6–7 · Teaching application

14 / 55 Canary deployment: limit exposure

Block 1 / Deployment & serving • Tomás



TO SAY

- An approved limited cohort uses the candidate before broader rollout.
- **Restrict exposure** to a defined cohort and scope.
- **Observe** execution, compatibility, coverage, and policy behavior.
- **Define stop rules** and a rehearsed recovery path.
- **Separate evidence:** short operational success cannot prove 30-day predictive quality.
- *At our bank:* The bank may authorize a small cohort to use M2 while retaining M1 for the remainder.
- *A canary controls risk; it is not automatically a valid business experiment.*

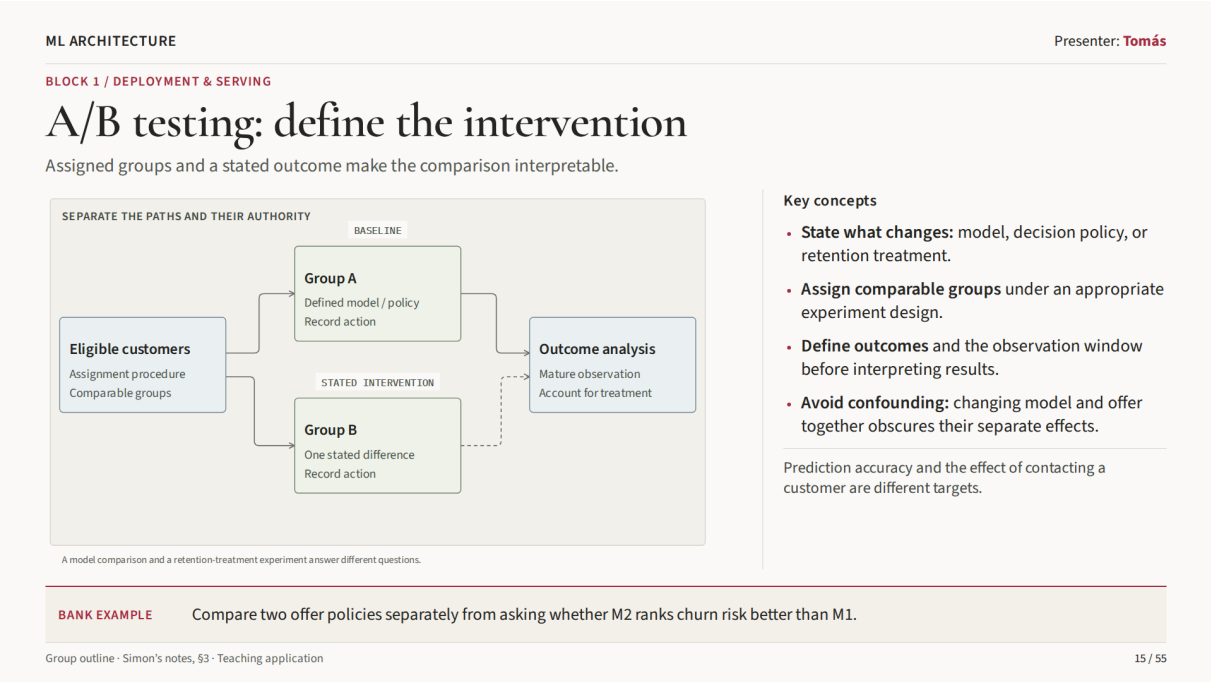
CUES

- Presenter: **Tomás**.
- Animated build: click 3 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
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Reading anchor: Breck, infrastructure test 6 · Bank teaching application

15 / 55 A/B testing: define the intervention

Block 1 / Deployment & serving • Tomás



TO SAY

- Assigned groups and a stated outcome make the comparison interpretable.
- **State what changes:** model, decision policy, or retention treatment.
- **Assign comparable groups** under an appropriate experiment design.
- **Define outcomes** and the observation window before interpreting results.
- **Avoid confounding:** changing model and offer together obscures their separate effects.
- *At our bank:* Compare two offer policies separately from asking whether M2 ranks churn risk better than M1.
- *Prediction accuracy and the effect of contacting a customer are different targets.*

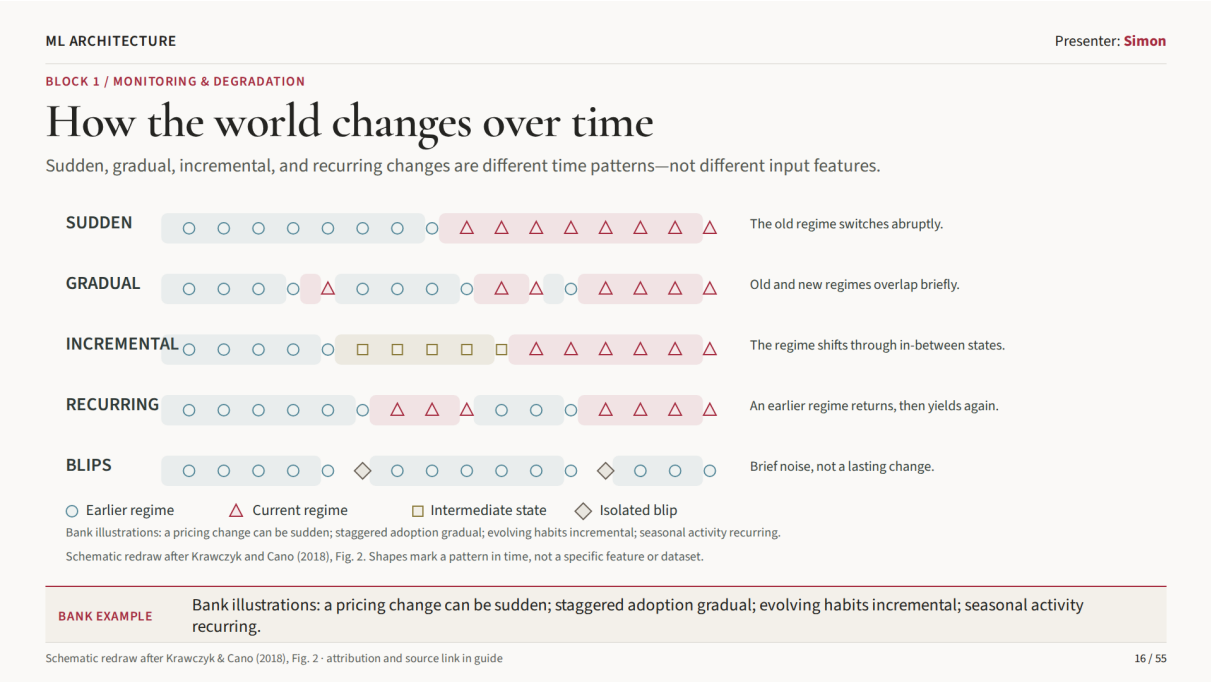
CUES

- Presenter: **Tomás**.
- Animated build: click 3 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

Reading anchor: Group outline · Simon's notes, §3 · Teaching application

16 / 55 How the world changes over time

Block 1 / Monitoring & degradation • Simon



TO SAY

- Sudden, gradual, incremental, and recurring changes are different time patterns—not different input features.
- **Sudden:** the old regime switches abruptly.
- **Gradual:** old and new regimes coexist during a transition.
- **Incremental / recurring:** intermediate change differs from a returning regime.
- **Blips:** isolated noise is not necessarily a lasting regime change.
- *At our bank:* Bank illustrations: a pricing change can be sudden; staggered adoption gradual; evolving habits incremental; seasonal activity recurring.
- *First ask how change unfolds over time. Then ask what changed: input data, outcome relationships, or data validity.*

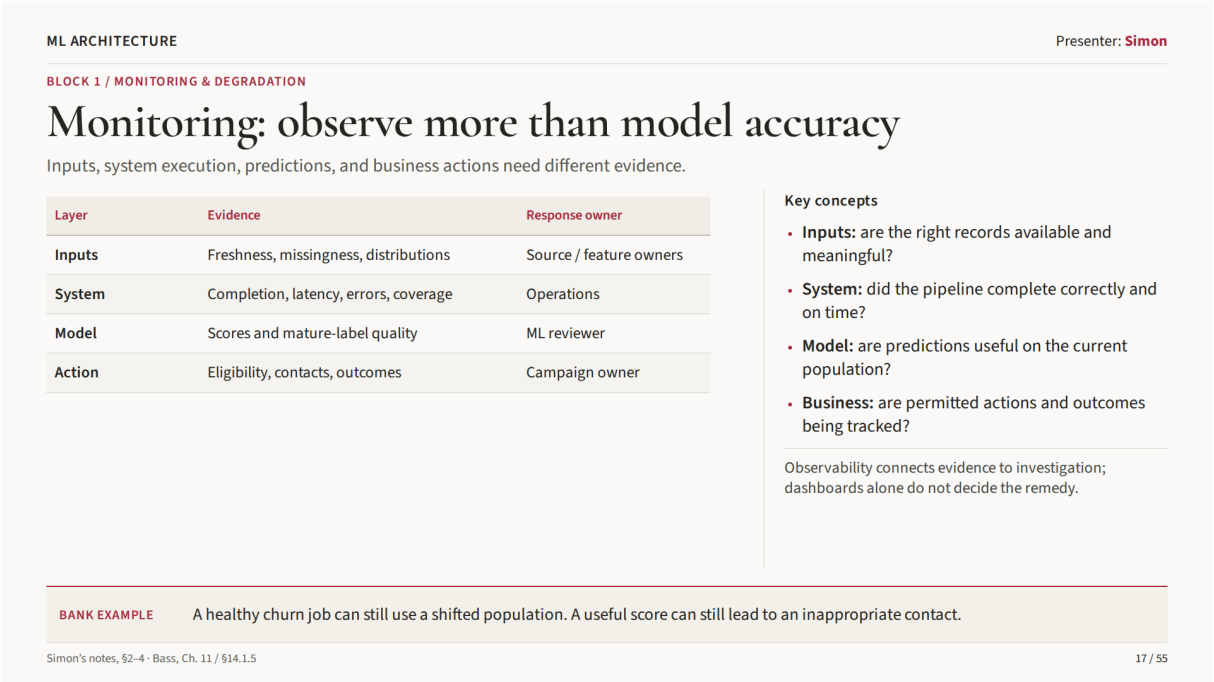
CUES

- New block: **Block 1 / Monitoring & degradation**.
- Presenter: **Simon**.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.
- Adaptation note on file for this slide; do not read aloud unless asked.

Reading anchor: Schematic redraw after Krawczyk & Cano (2018), Fig. 2 · attribution and source link in guide

17 / 55 Monitoring: observe more than model accuracy

Block 1 / Monitoring & degradation • Simon



TO SAY

- Inputs, system execution, predictions, and business actions need different evidence.
- **Inputs:** are the right records available and meaningful?
- **System:** did the pipeline complete correctly and on time?
- **Model:** are predictions useful on the current population?
- **Business:** are permitted actions and outcomes being tracked?
- *At our bank:* A healthy churn job can still use a shifted population. A useful score can still lead to an inappropriate contact.
- *Observability connects evidence to investigation; dashboards alone do not decide the remedy.*

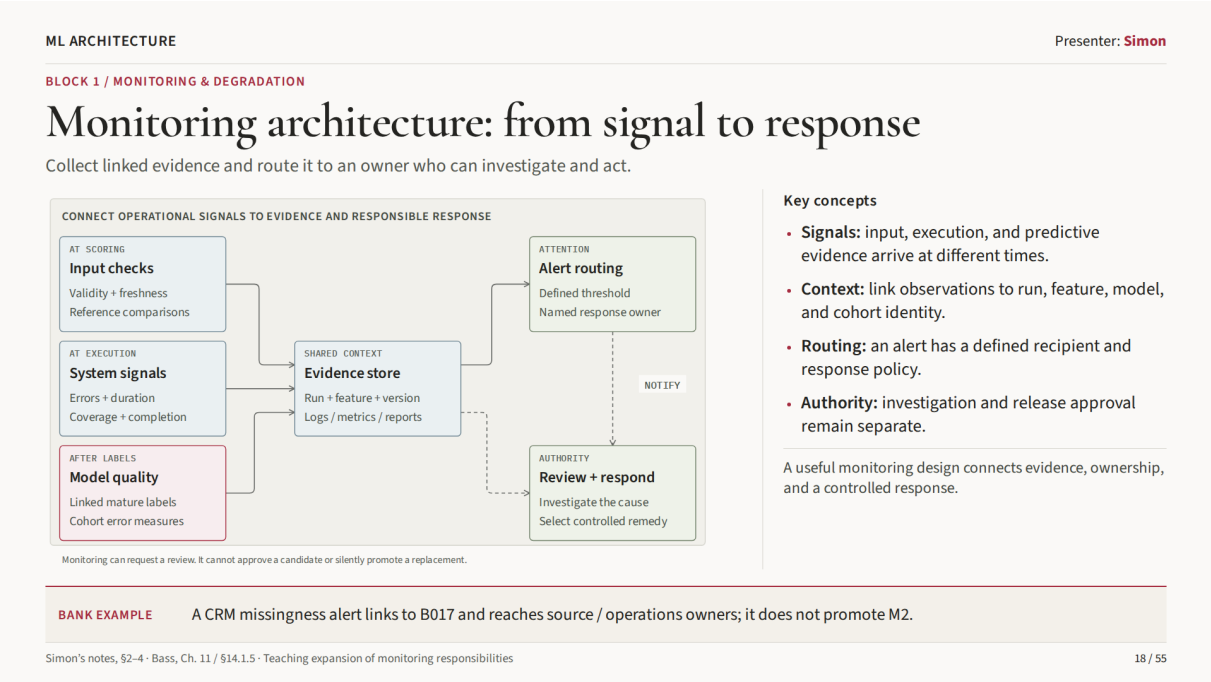
CUES

- Presenter: **Simon**.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

Reading anchor: Simon's notes, §2–4 · Bass, Ch. 11 / §14.1.5

18 / 55 Monitoring architecture: from signal to response

Block 1 / Monitoring & degradation • Simon



TO SAY

- Collect linked evidence and route it to an owner who can investigate and act.
- **Signals:** input, execution, and predictive evidence arrive at different times.
- **Context:** link observations to run, feature, model, and cohort identity.
- **Routing:** an alert has a defined recipient and response policy.
- **Authority:** investigation and release approval remain separate.
- *At our bank:* A CRM missingness alert links to B017 and reaches source / operations owners; it does not promote M2.
- *A useful monitoring design connects evidence, ownership, and a controlled response.*

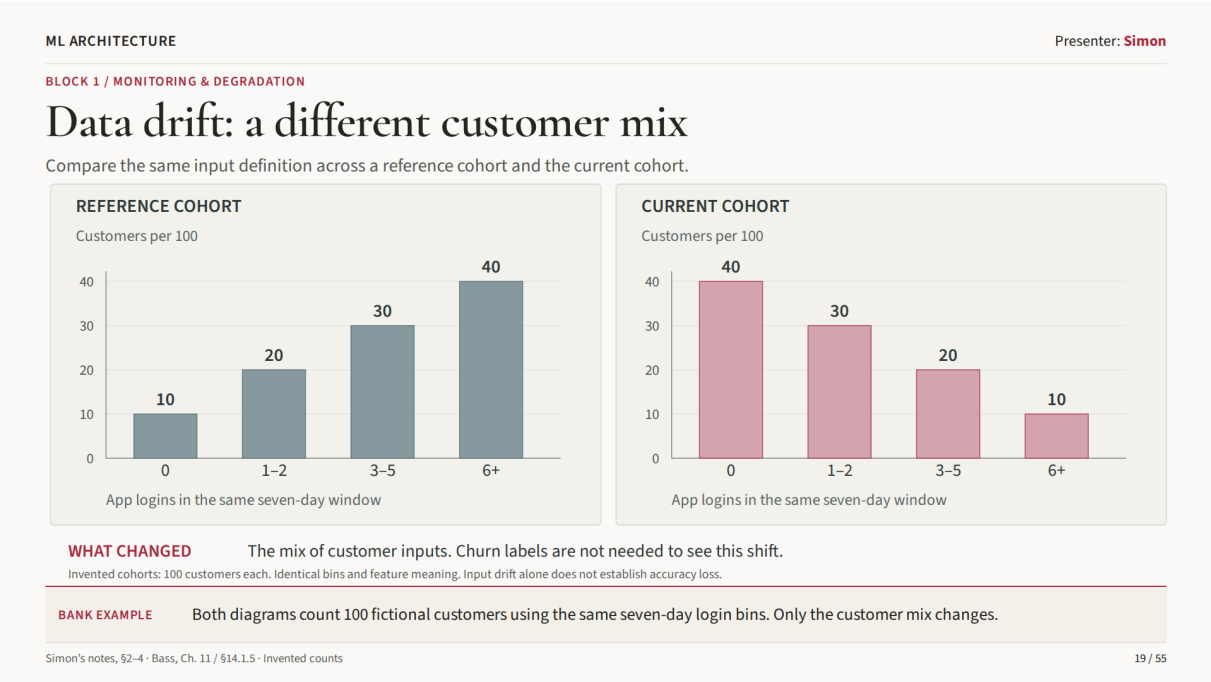
CUES

- Presenter: **Simon**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

Reading anchor: Simon's notes, §2–4 · Bass, Ch. 11 / §14.1.5 · Teaching expansion of monitoring responsibilities

19 / 55 Data drift: a different customer mix

Block 1 / Monitoring & degradation • Simon



TO SAY

- Compare the same input definition across a reference cohort and the current cohort.
- **What changes:** the distribution of valid model inputs.
- **Bank example:** a campaign attracts more low-activity customers.
- **Available now:** compare inputs without waiting for churn labels.
- **Limit:** a changed distribution alone does not prove lost accuracy.
- *At our bank:* Both diagrams count 100 fictional customers using the same seven-day login bins. Only the customer mix changes.
- *Choose a meaningful reference, comparable cohorts, and a defined response.*

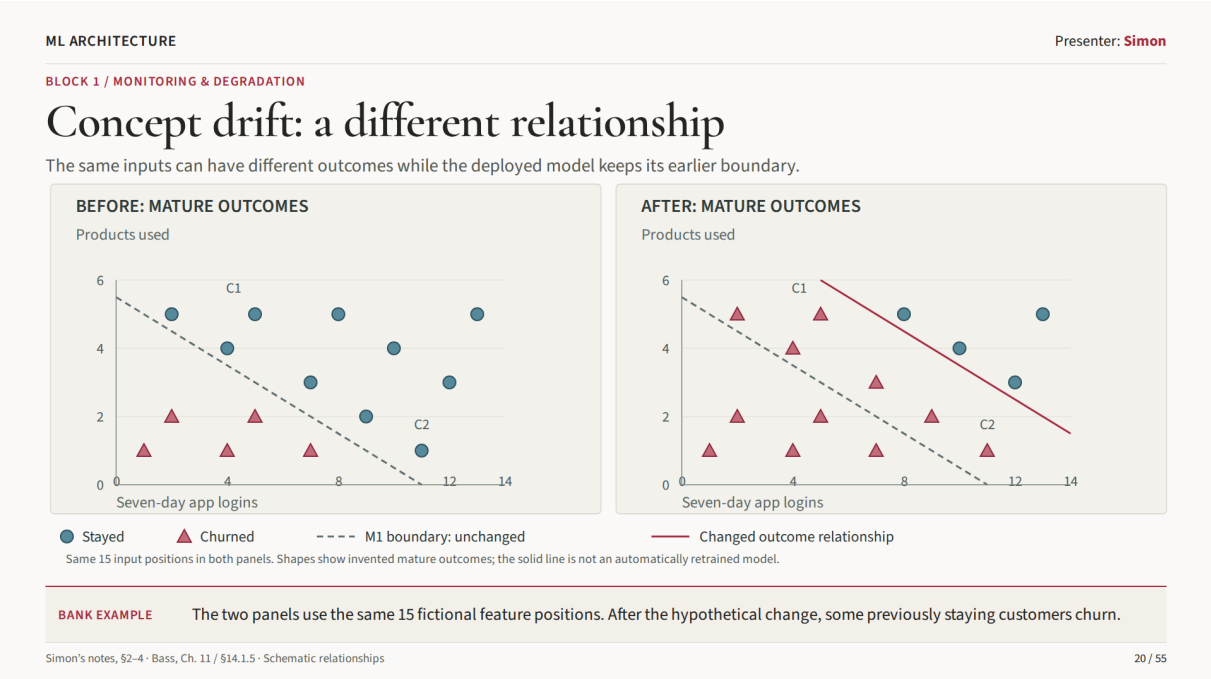
CUES

- Presenter: **Simon**.
- Animated build: click 1 time to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

Reading anchor: Simon's notes, §2–4 · Bass, Ch. 11 / §14.1.5 · Invented counts

20 / 55 Concept drift: a different relationship

Block 1 / Monitoring & degradation • Simon



TO SAY

- The same inputs can have different outcomes while the deployed model keeps its earlier boundary.
- **What changes:** churn risk given the same valid input information.
- **Bank example:** new fees make active customers leave more often.
- **Evidence:** link predictions to mature, reliable churn outcomes.
- **Model status:** M1 stays fixed until a separately evaluated release changes it.
- *At our bank:* The two panels use the same 15 fictional feature positions. After the hypothetical change, some previously staying customers churn.
- *A missing field is not concept drift; a statistical alarm is not a retraining approval.*

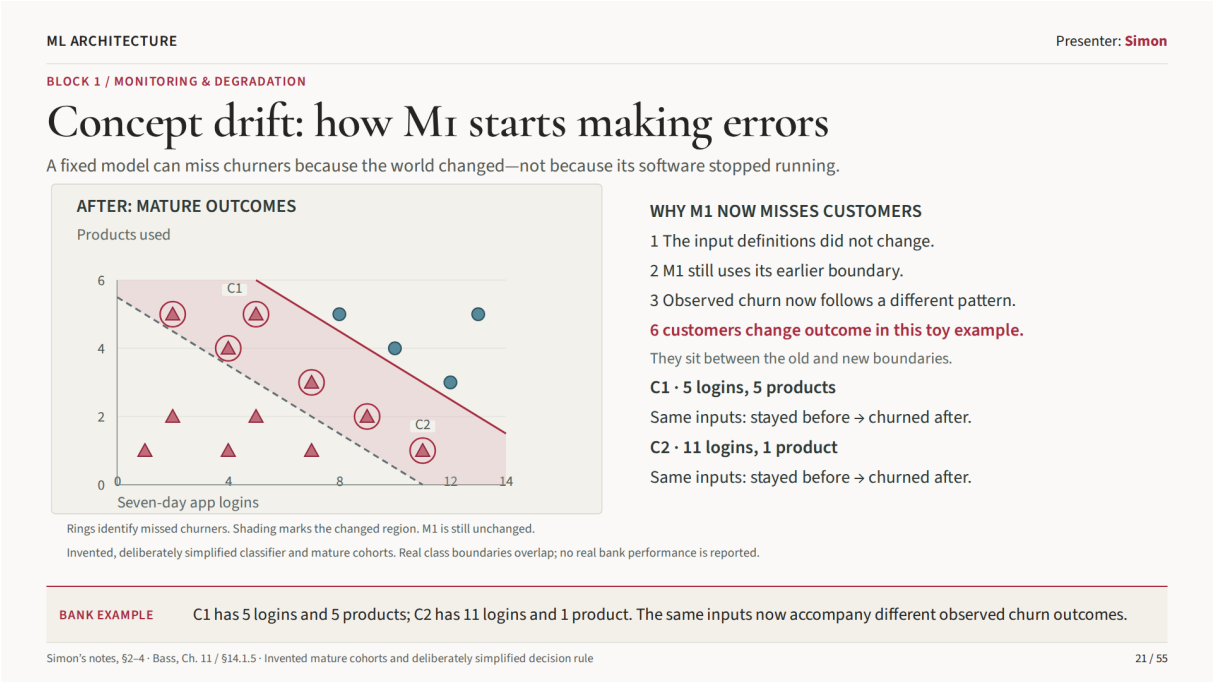
CUES

- Presenter: **Simon**.
- Animated build: click 1 time to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

Reading anchor: Simon's notes, §2–4 · Bass, Ch. 11 / §14.1.5 · Schematic relationships

21 / 55 Concept drift: how M1 starts making errors

Block 1 / Monitoring & degradation • Simon



TO SAY

- A fixed model can miss churners because the world changed—not because its software stopped running.
- **Inputs stay fixed:** the two cohorts use identical feature positions.
- **M1 stays fixed:** its earlier decision boundary is unchanged.
- **Outcomes change:** a different relationship creates missed churners.
- **Evidence:** the example uses mature labels, not outcomes already known on scoring day.
- *At our bank:* C1 has 5 logins and 5 products; C2 has 11 logins and 1 product. The same inputs now accompany different observed churn outcomes.
- *Changed labels reveal the mismatch. Investigate causes, evaluate candidates, and use controlled release rather than automatic promotion.*

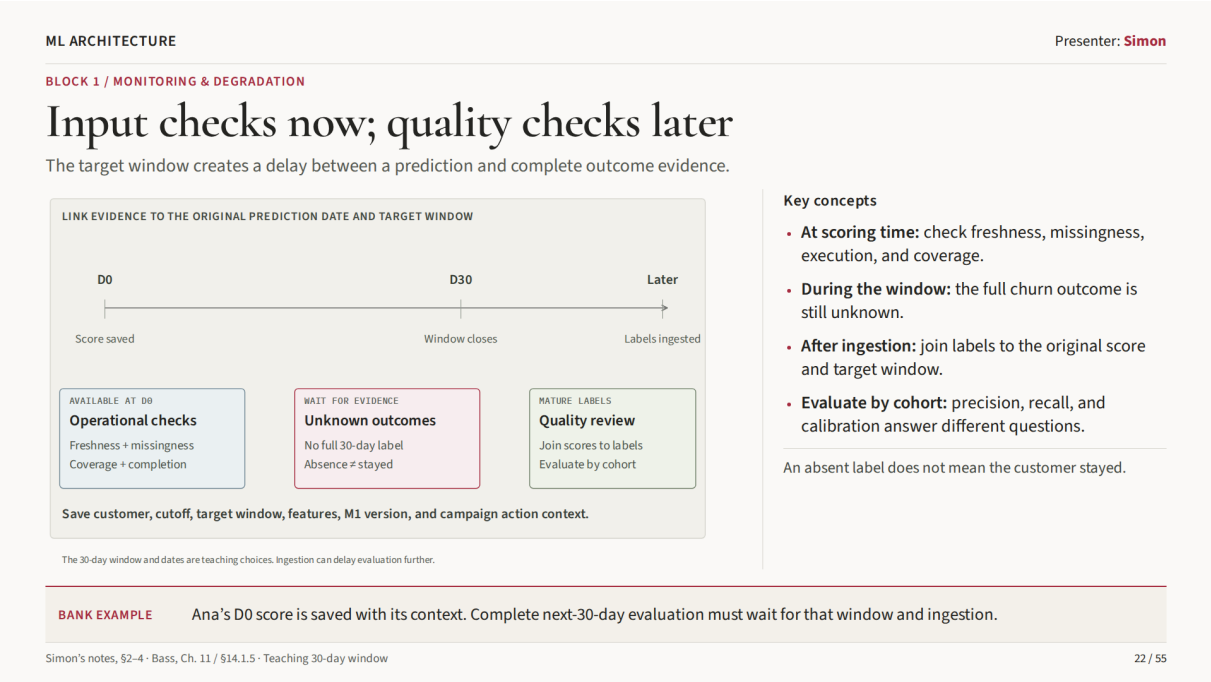
CUES

- Presenter: **Simon**.
- Animated build: click 1 time to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.
- Adaptation note on file for this slide; do not read aloud unless asked.

Reading anchor: Simon's notes, §2–4 · Bass, Ch. 11 / §14.1.5 · Invented mature cohorts and deliberately simplified decision rule

22 / 55 Input checks now; quality checks later

Block 1 / Monitoring & degradation • Simon



TO SAY

- The target window creates a delay between a prediction and complete outcome evidence.
- **At scoring time:** check freshness, missingness, execution, and coverage.
- **During the window:** the full churn outcome is still unknown.
- **After ingestion:** join labels to the original score and target window.
- **Evaluate by cohort:** precision, recall, and calibration answer different questions.
- *At our bank:* Ana's D0 score is saved with its context. Complete next-30-day evaluation must wait for that window and ingestion.
- *An absent label does not mean the customer stayed.*

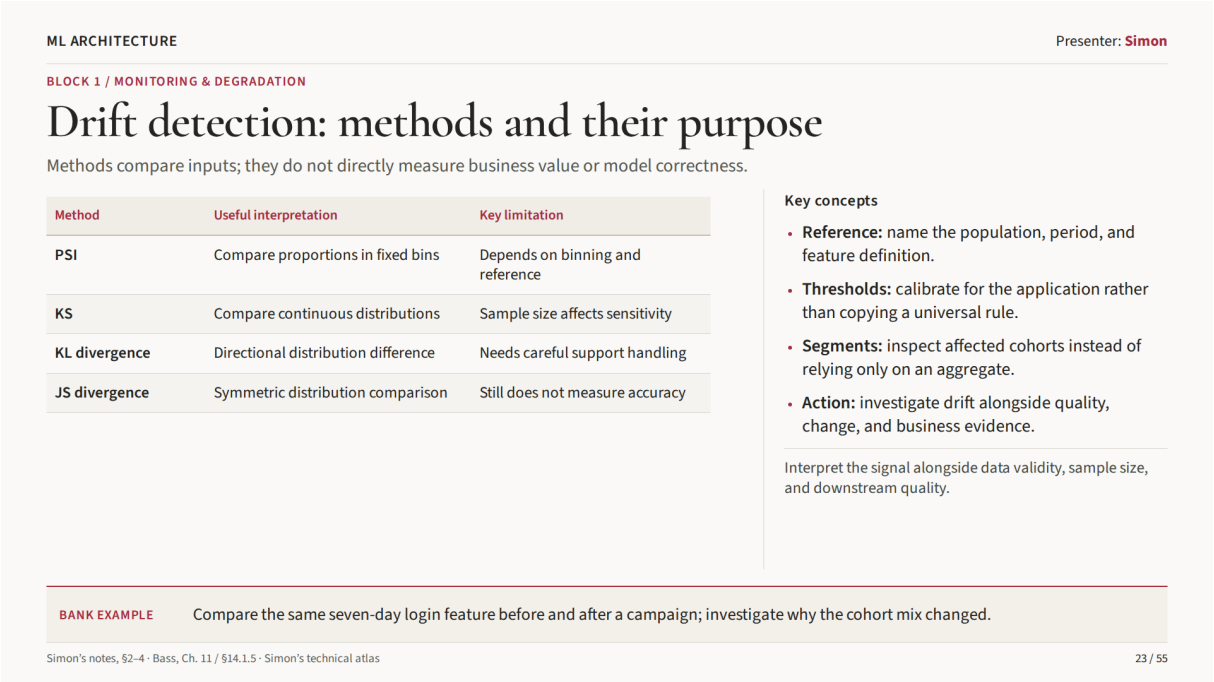
CUES

- Presenter: **Simon**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

Reading anchor: Simon's notes, §2–4 · Bass, Ch. 11 / §14.1.5 · Teaching 30-day window

23 / 55 Drift detection: methods and their purpose

Block 1 / Monitoring & degradation • Simon



TO SAY

- Methods compare inputs; they do not directly measure business value or model correctness.
- **Reference:** name the population, period, and feature definition.
- **Thresholds:** calibrate for the application rather than copying a universal rule.
- **Segments:** inspect affected cohorts instead of relying only on an aggregate.
- **Action:** investigate drift alongside quality, change, and business evidence.
- *At our bank:* Compare the same seven-day login feature before and after a campaign; investigate why the cohort mix changed.
- *Interpret the signal alongside data validity, sample size, and downstream quality.*

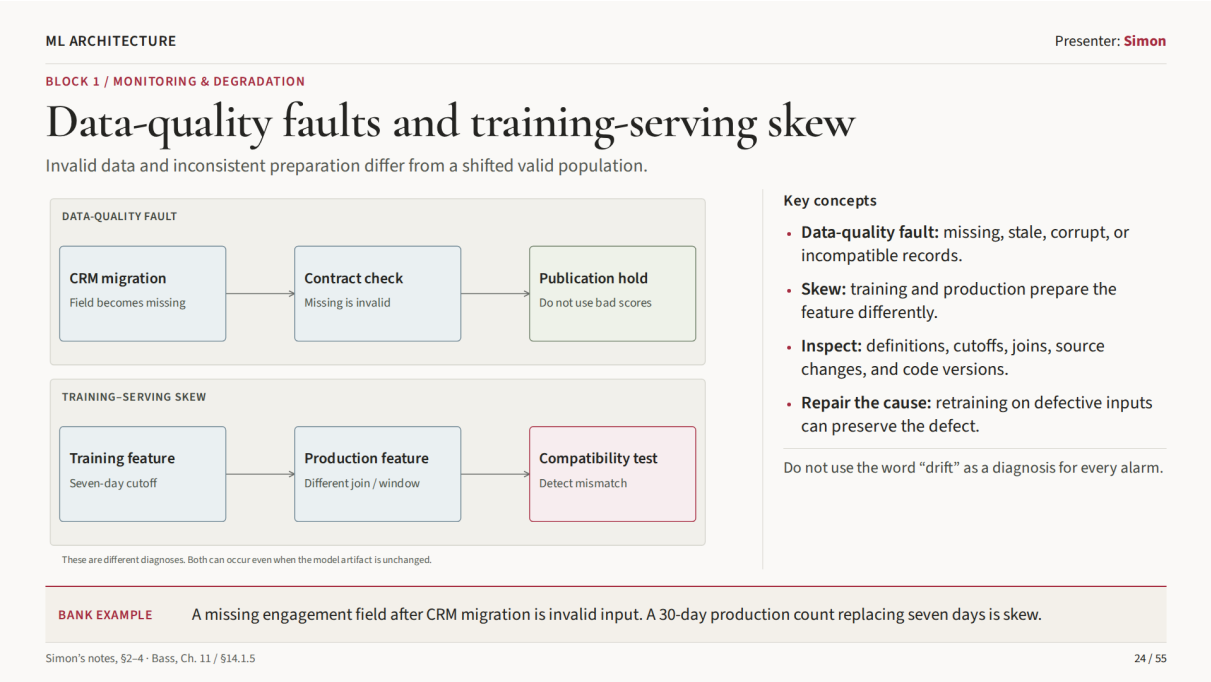
CUES

- Presenter: **Simon**.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

Reading anchor: Simon's notes, §2–4 · Bass, Ch. 11 / §14.1.5 · Simon's technical atlas

24 / 55 Data-quality faults and training-serving skew

Block 1 / Monitoring & degradation • Simon



TO SAY

- Invalid data and inconsistent preparation differ from a shifted valid population.
- **Data-quality fault:** missing, stale, corrupt, or incompatible records.
- **Skew:** training and production prepare the feature differently.
- **Inspect:** definitions, cutoffs, joins, source changes, and code versions.
- **Repair the cause:** retraining on defective inputs can preserve the defect.
- *At our bank:* A missing engagement field after CRM migration is invalid input. A 30-day production count replacing seven days is skew.
- *Do not use the word “drift” as a diagnosis for every alarm.*

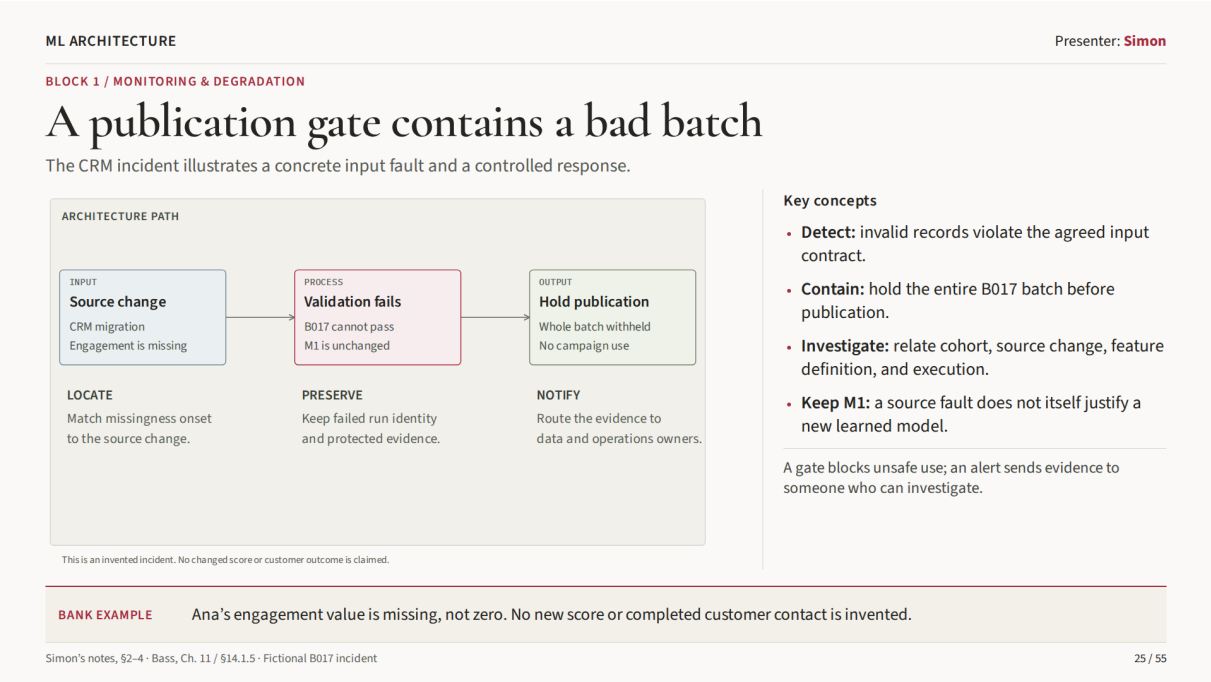
CUES

- Presenter: **Simon**.
- Animated build: click 3 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

Reading anchor: Simon’s notes, §2–4 · Bass, Ch. 11 / §14.1.5

25 / 55 A publication gate contains a bad batch

Block 1 / Monitoring & degradation • Simon



TO SAY

- The CRM incident illustrates a concrete input fault and a controlled response.
- **Detect:** invalid records violate the agreed input contract.
- **Contain:** hold the entire B017 batch before publication.
- **Investigate:** relate cohort, source change, feature definition, and execution.
- **Keep M1:** a source fault does not itself justify a new learned model.
- *At our bank:* Ana's engagement value is missing, not zero. No new score or completed customer contact is invented.
- *A gate blocks unsafe use; an alert sends evidence to someone who can investigate.*

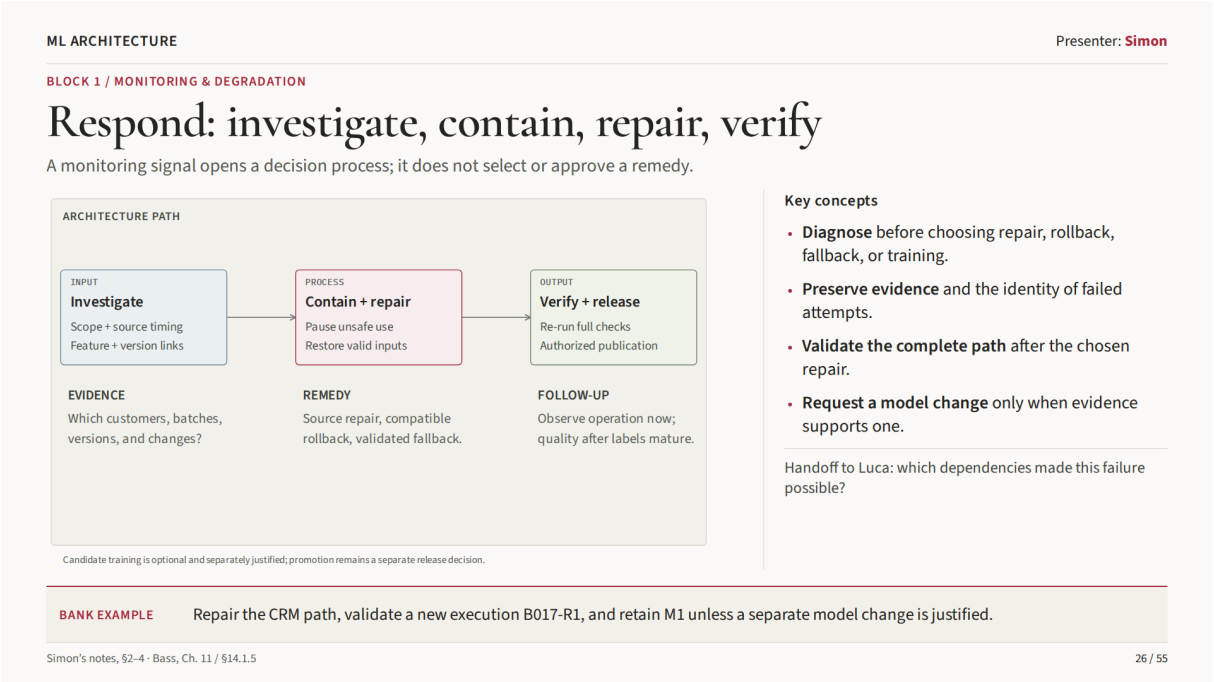
CUES

- Presenter: **Simon**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

Reading anchor: Simon's notes, §2–4 · Bass, Ch. 11 / §14.1.5 · Fictional B017 incident

26 / 55 Respond: investigate, contain, repair, verify

Block 1 / Monitoring & degradation • Simon



TO SAY

- A monitoring signal opens a decision process; it does not select or approve a remedy.
- **Diagnose** before choosing repair, rollback, fallback, or training.
- **Preserve evidence** and the identity of failed attempts.
- **Validate the complete path** after the chosen repair.
- **Request a model change** only when evidence supports one.
- *At our bank:* Repair the CRM path, validate a new execution B017-R1, and retain M1 unless a separate model change is justified.
- *Handoff to Luca: which dependencies made this failure possible?*

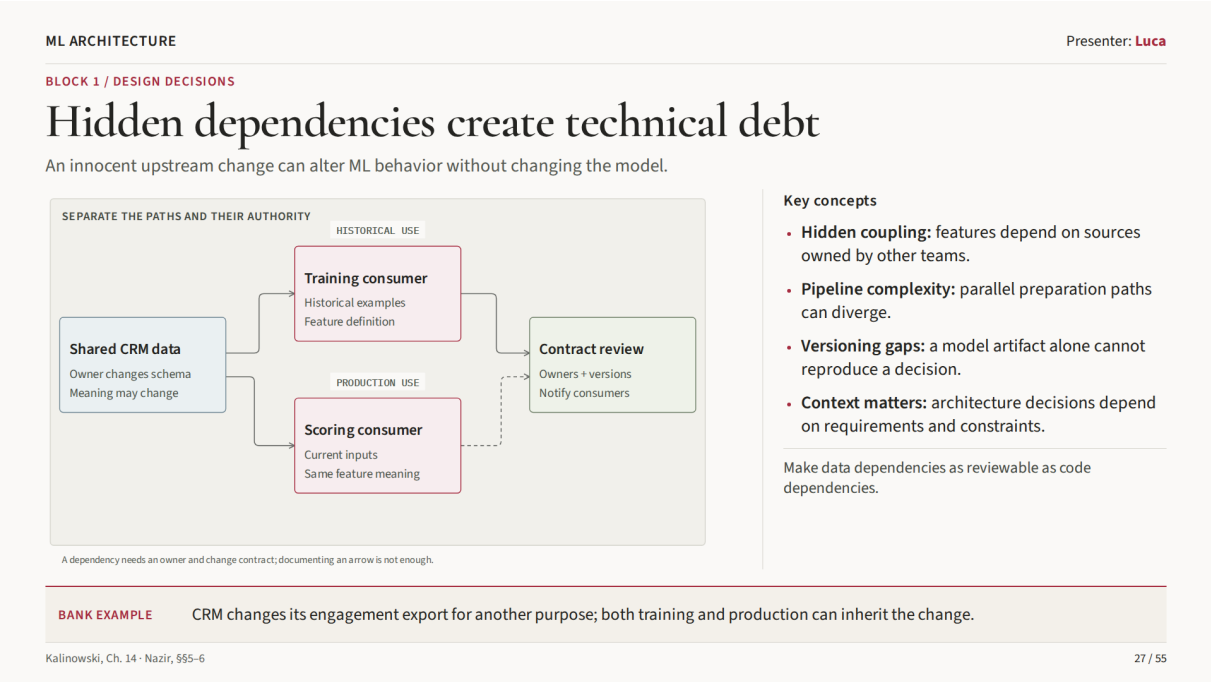
CUES

- Presenter: **Simon**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

Reading anchor: Simon's notes, §2–4 · Bass, Ch. 11 / §14.1.5

27 / 55 Hidden dependencies create technical debt

Block 1 / Design decisions • Luca



TO SAY

- An innocent upstream change can alter ML behavior without changing the model.
- **Hidden coupling:** features depend on sources owned by other teams.
- **Pipeline complexity:** parallel preparation paths can diverge.
- **Versioning gaps:** a model artifact alone cannot reproduce a decision.
- **Context matters:** architecture decisions depend on requirements and constraints.
- *At our bank:* CRM changes its engagement export for another purpose; both training and production can inherit the change.
- *Make data dependencies as reviewable as code dependencies.*

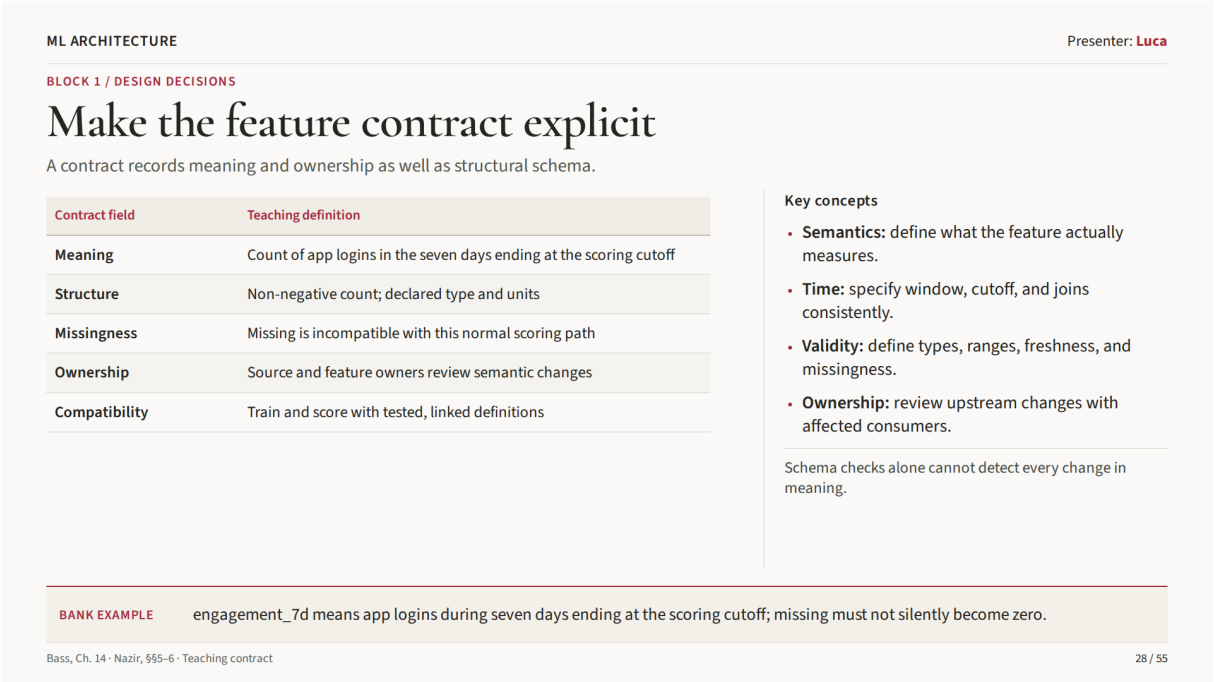
CUES

- New block: **Block 1 / Design decisions**.
- Presenter: **Luca**.
- Animated build: click 3 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

Reading anchor: Kalinowski, Ch. 14 · Nazir, §§5–6

28 / 55 Make the feature contract explicit

Block 1 / Design decisions • Luca



TO SAY

- A contract records meaning and ownership as well as structural schema.
- **Semantics:** define what the feature actually measures.
- **Time:** specify window, cutoff, and joins consistently.
- **Validity:** define types, ranges, freshness, and missingness.
- **Ownership:** review upstream changes with affected consumers.
- *At our bank:* engagement_7d means app logins during seven days ending at the scoring cutoff; missing must not silently become zero.
- *Schema checks alone cannot detect every change in meaning.*

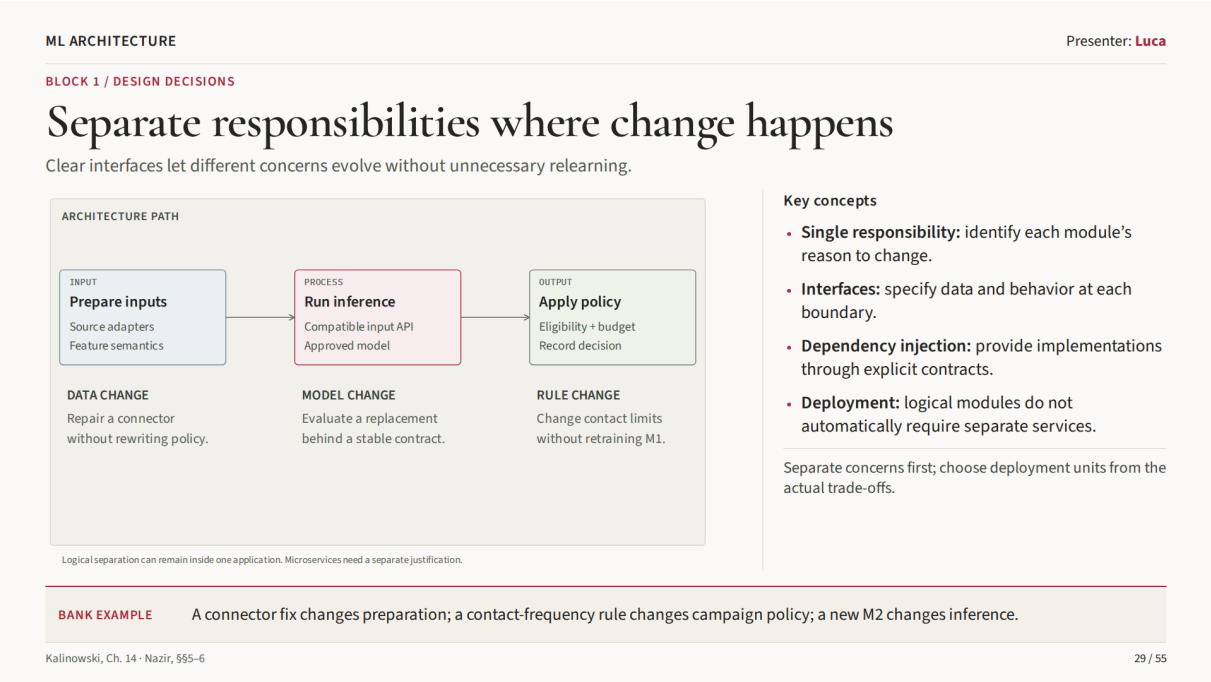
CUES

- Presenter: **Luca**.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

Reading anchor: Bass, Ch. 14 · Nazir, §§5–6 · Teaching contract

29 / 55 Separate responsibilities where change happens

Block 1 / Design decisions • Luca



TO SAY

- Clear interfaces let different concerns evolve without unnecessary relearning.
- **Single responsibility:** identify each module's reason to change.
- **Interfaces:** specify data and behavior at each boundary.
- **Dependency injection:** provide implementations through explicit contracts.
- **Deployment:** logical modules do not automatically require separate services.
- *At our bank:* A connector fix changes preparation; a contact-frequency rule changes campaign policy; a new M2 changes inference.
- *Separate concerns first; choose deployment units from the actual trade-offs.*

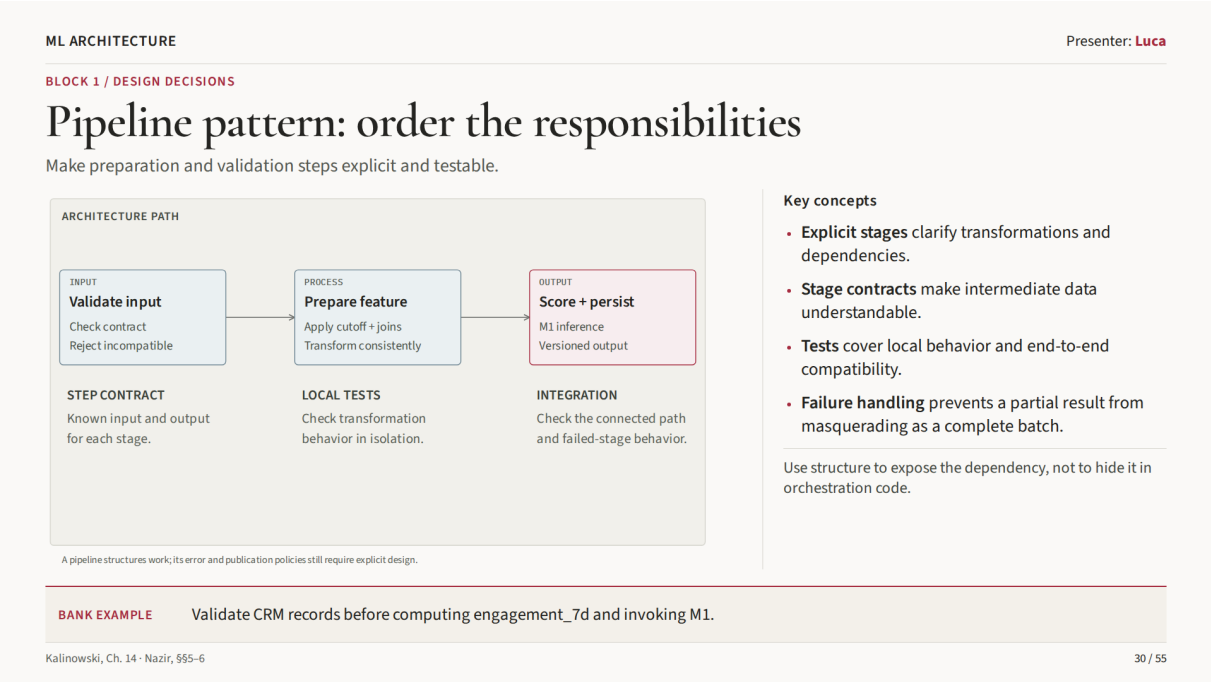
CUES

- Presenter: **Luca**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

Reading anchor: Kalinowski, Ch. 14 · Nazir, §§5–6

30 / 55 Pipeline pattern: order the responsibilities

Block 1 / Design decisions • Luca



TO SAY

- Make preparation and validation steps explicit and testable.
- **Explicit stages** clarify transformations and dependencies.
- **Stage contracts** make intermediate data understandable.
- **Tests** cover local behavior and end-to-end compatibility.
- **Failure handling** prevents a partial result from masquerading as a complete batch.
- *At our bank:* Validate CRM records before computing engagement_7d and invoking M1.
- *Use structure to expose the dependency, not to hide it in orchestration code.*

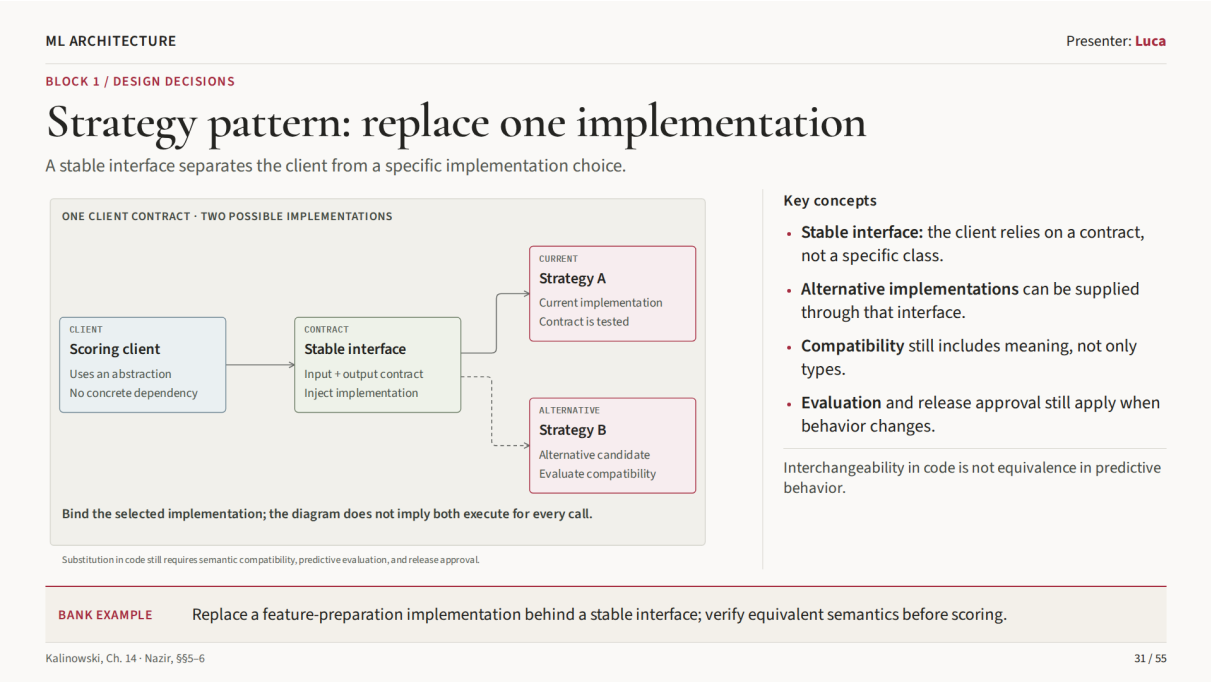
CUES

- Presenter: **Luca**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

Reading anchor: Kalinowski, Ch. 14 · Nazir, §§5–6

31 / 55 Strategy pattern: replace one implementation

Block 1 / Design decisions • Luca



TO SAY

- A stable interface separates the client from a specific implementation choice.
- **Stable interface:** the client relies on a contract, not a specific class.
- **Alternative implementations** can be supplied through that interface.
- **Compatibility** still includes meaning, not only types.
- **Evaluation** and release approval still apply when behavior changes.
- *At our bank:* Replace a feature-preparation implementation behind a stable interface; verify equivalent semantics before scoring.
- *Interchangeability in code is not equivalence in predictive behavior.*

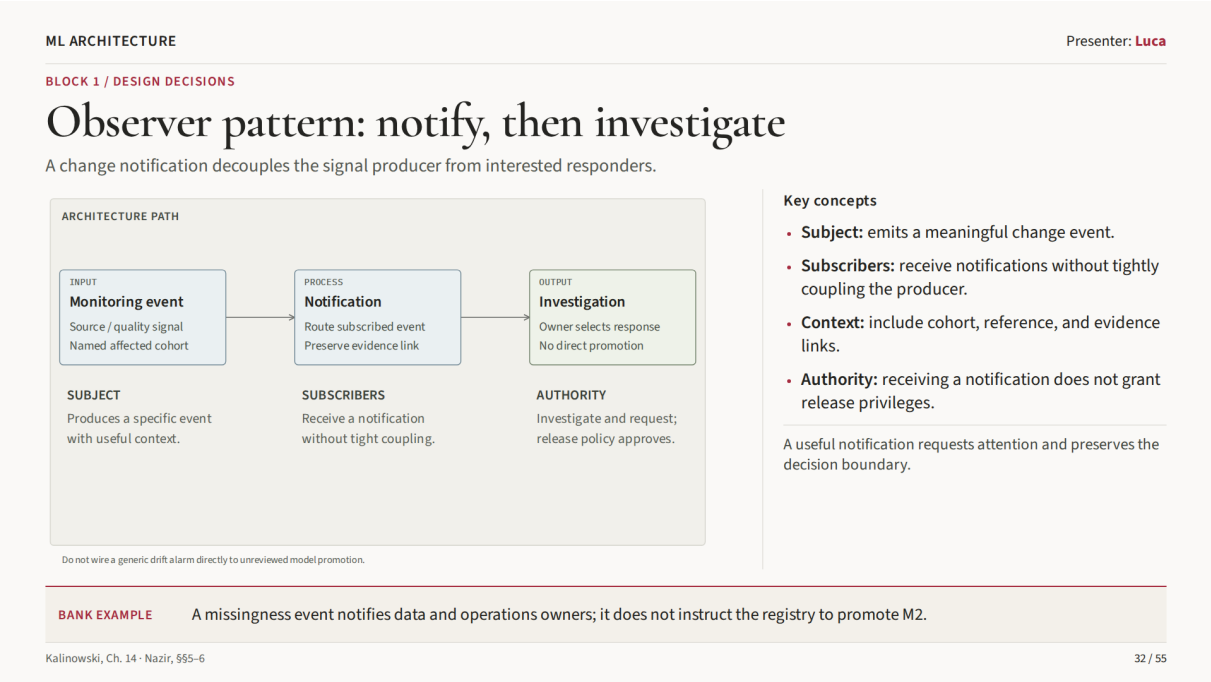
CUES

- Presenter: **Luca**.
- Animated build: click 3 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

Reading anchor: Kalinowski, Ch. 14 · Nazir, §§5–6

32 / 55 Observer pattern: notify, then investigate

Block 1 / Design decisions • Luca



TO SAY

- A change notification decouples the signal producer from interested responders.
- **Subject:** emits a meaningful change event.
- **Subscribers:** receive notifications without tightly coupling the producer.
- **Context:** include cohort, reference, and evidence links.
- **Authority:** receiving a notification does not grant release privileges.
- *At our bank:* A missingness event notifies data and operations owners; it does not instruct the registry to promote M2.
- *A useful notification requests attention and preserves the decision boundary.*

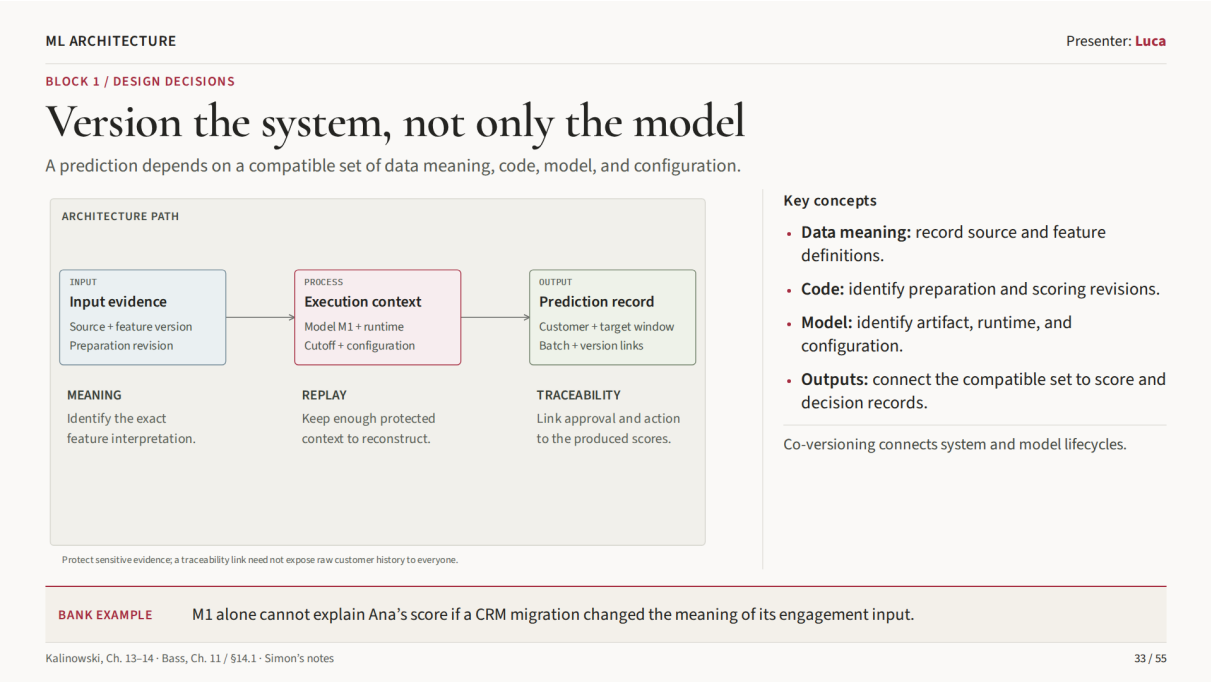
CUES

- Presenter: **Luca**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

Reading anchor: Kalinowski, Ch. 14 · Nazir, §§5–6

33 / 55 Version the system, not only the model

Block 1 / Design decisions • Luca



TO SAY

- A prediction depends on a compatible set of data meaning, code, model, and configuration.
- **Data meaning:** record source and feature definitions.
- **Code:** identify preparation and scoring revisions.
- **Model:** identify artifact, runtime, and configuration.
- **Outputs:** connect the compatible set to score and decision records.
- *At our bank:* M1 alone cannot explain Ana's score if a CRM migration changed the meaning of its engagement input.
- *Co-versioning connects system and model lifecycles.*

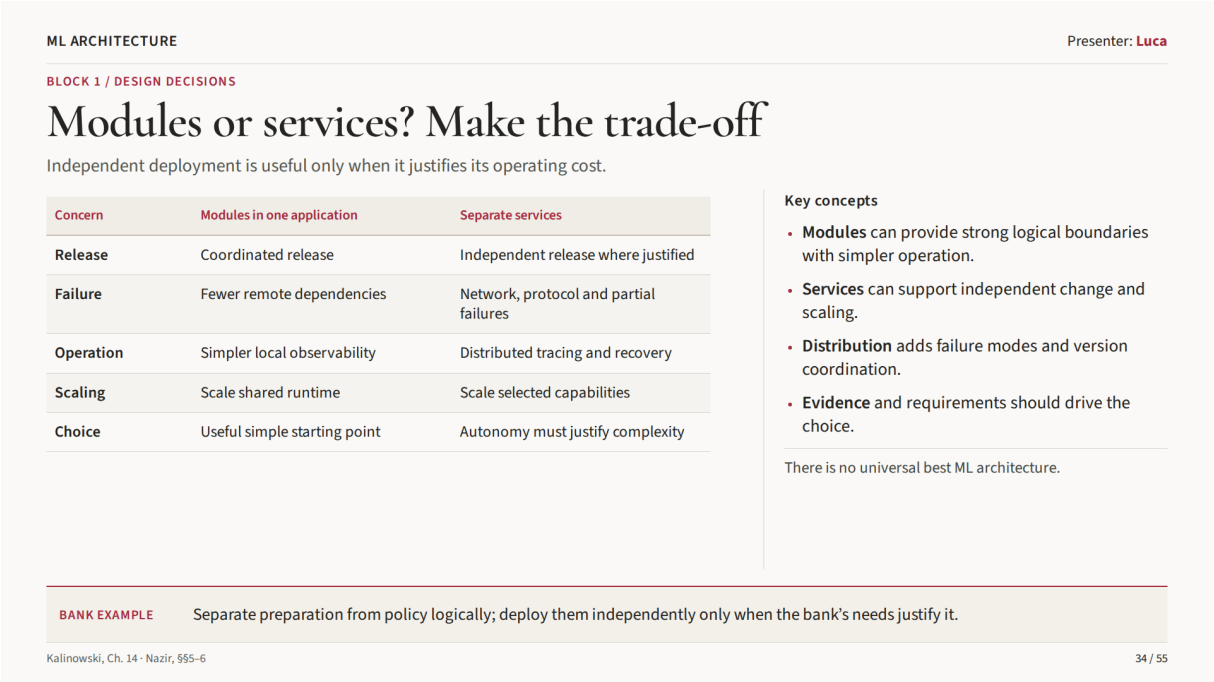
CUES

- Presenter: **Luca**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

Reading anchor: Kalinowski, Ch. 13–14 · Bass, Ch. 11 / §14.1 · Simon's notes

34 / 55 Modules or services? Make the trade-off

Block 1 / Design decisions • Luca



TO SAY

- Independent deployment is useful only when it justifies its operating cost.
- **Modules** can provide strong logical boundaries with simpler operation.
- **Services** can support independent change and scaling.
- **Distribution** adds failure modes and version coordination.
- **Evidence** and requirements should drive the choice.
- *At our bank:* Separate preparation from policy logically; deploy them independently only when the bank's needs justify it.
- *There is no universal best ML architecture.*

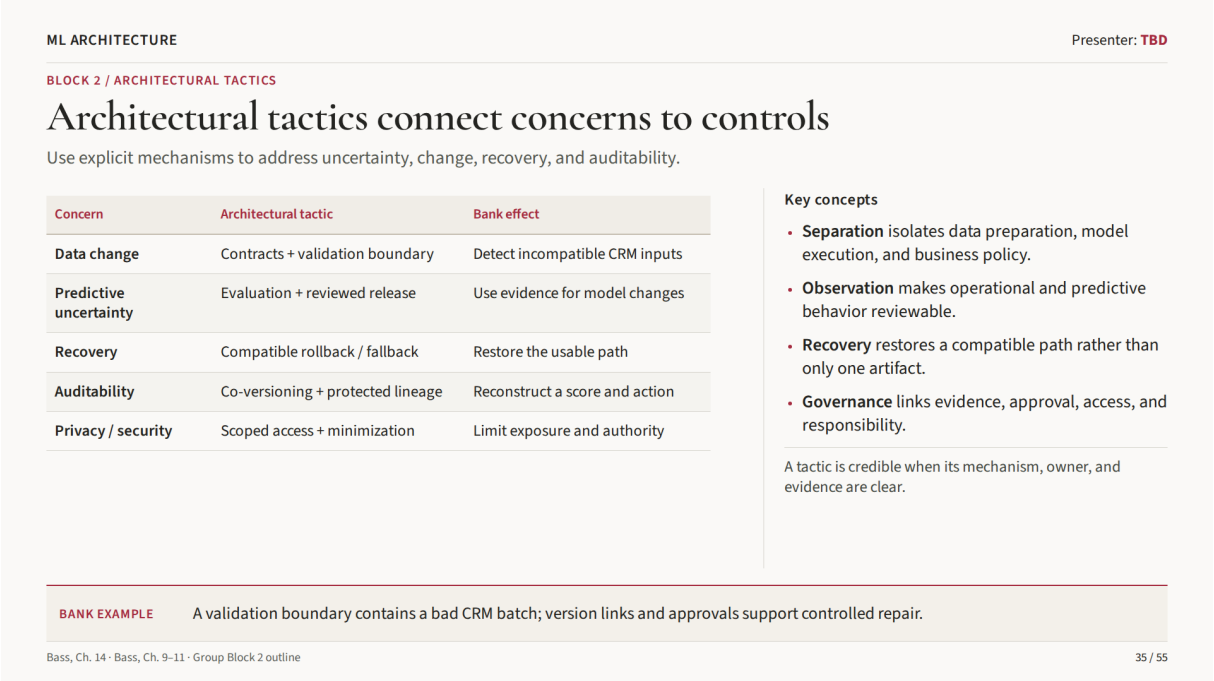
CUES

- Presenter: **Luca**.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

Reading anchor: Kalinowski, Ch. 14 · Nazir, §§5–6

35 / 55 Architectural tactics connect concerns to controls

Block 2 / Architectural tactics • TBD



TO SAY

- Use explicit mechanisms to address uncertainty, change, recovery, and auditability.
- **Separation** isolates data preparation, model execution, and business policy.
- **Observation** makes operational and predictive behavior reviewable.
- **Recovery** restores a compatible path rather than only one artifact.
- **Governance** links evidence, approval, access, and responsibility.
- *At our bank:* A validation boundary contains a bad CRM batch; version links and approvals support controlled repair.
- *A tactic is credible when its mechanism, owner, and evidence are clear.*

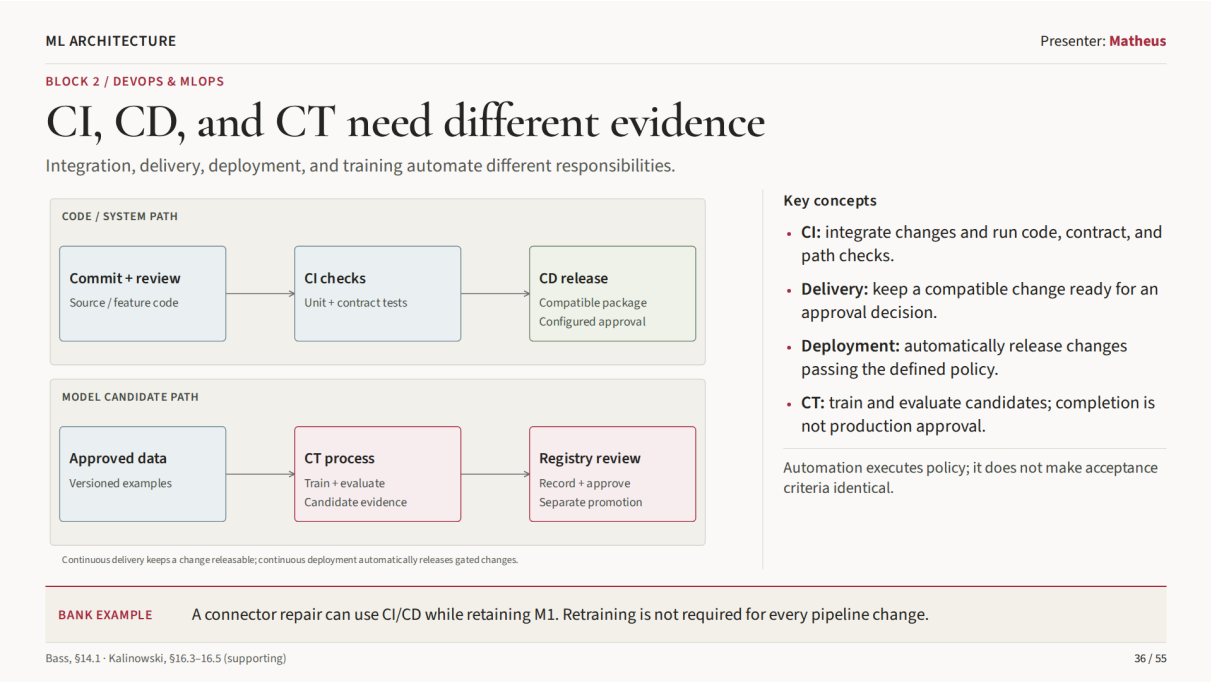
CUES

- New block: **Block 2 / Architectural tactics**.
- Presenter: **TBD**.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

Reading anchor: Bass, Ch. 14 · Bass, Ch. 9–11 · Group Block 2 outline

36 / 55 CI, CD, and CT need different evidence

Block 2 / DevOps & MLOps • Matheus



TO SAY

- Integration, delivery, deployment, and training automate different responsibilities.
- **CI:** integrate changes and run code, contract, and path checks.
- **Delivery:** keep a compatible change ready for an approval decision.
- **Deployment:** automatically release changes passing the defined policy.
- **CT:** train and evaluate candidates; completion is not production approval.
- *At our bank:* A connector repair can use CI/CD while retaining M1. Retraining is not required for every pipeline change.
- *Automation executes policy; it does not make acceptance criteria identical.*

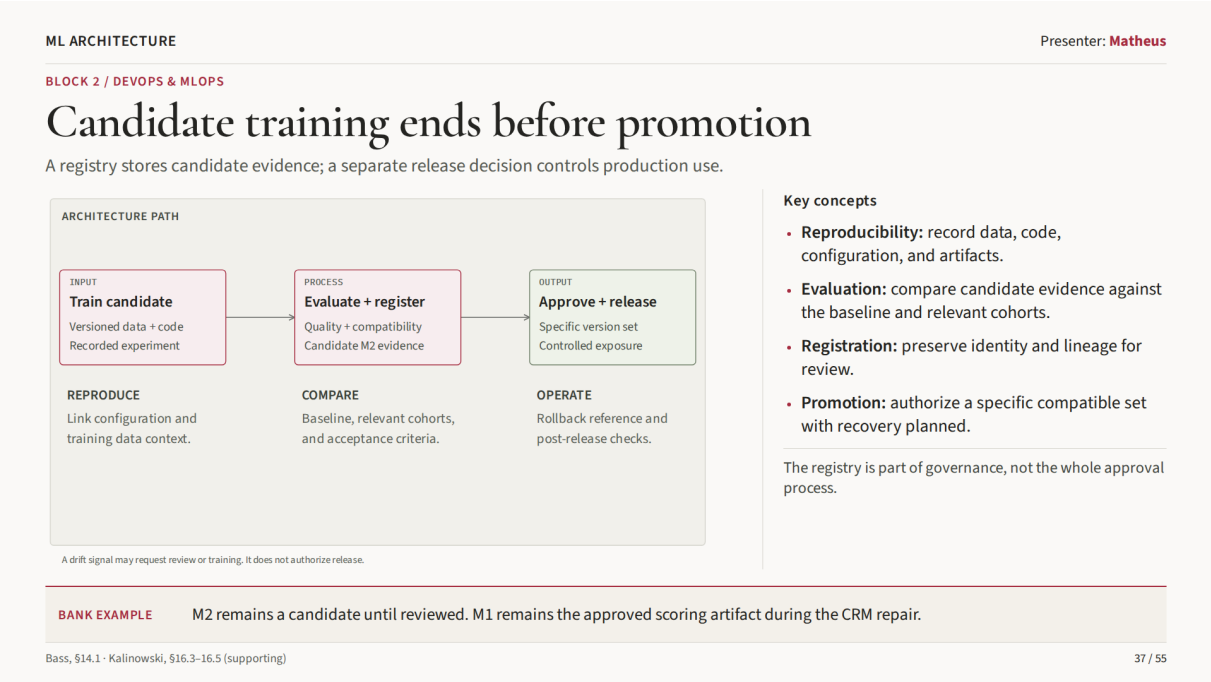
CUES

- New block: **Block 2 / DevOps & MLOps**.
- Presenter: **Matheus**.
- Animated build: click 3 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

Reading anchor: Bass, §14.1 · Kalinowski, §16.3–16.5 (supporting)

37 / 55 Candidate training ends before promotion

Block 2 / DevOps & MLOps • Matheus



TO SAY

- A registry stores candidate evidence; a separate release decision controls production use.
- **Reproducibility:** record data, code, configuration, and artifacts.
- **Evaluation:** compare candidate evidence against the baseline and relevant cohorts.
- **Registration:** preserve identity and lineage for review.
- **Promotion:** authorize a specific compatible set with recovery planned.
- *At our bank:* M2 remains a candidate until reviewed. M1 remains the approved scoring artifact during the CRM repair.
- *The registry is part of governance, not the whole approval process.*

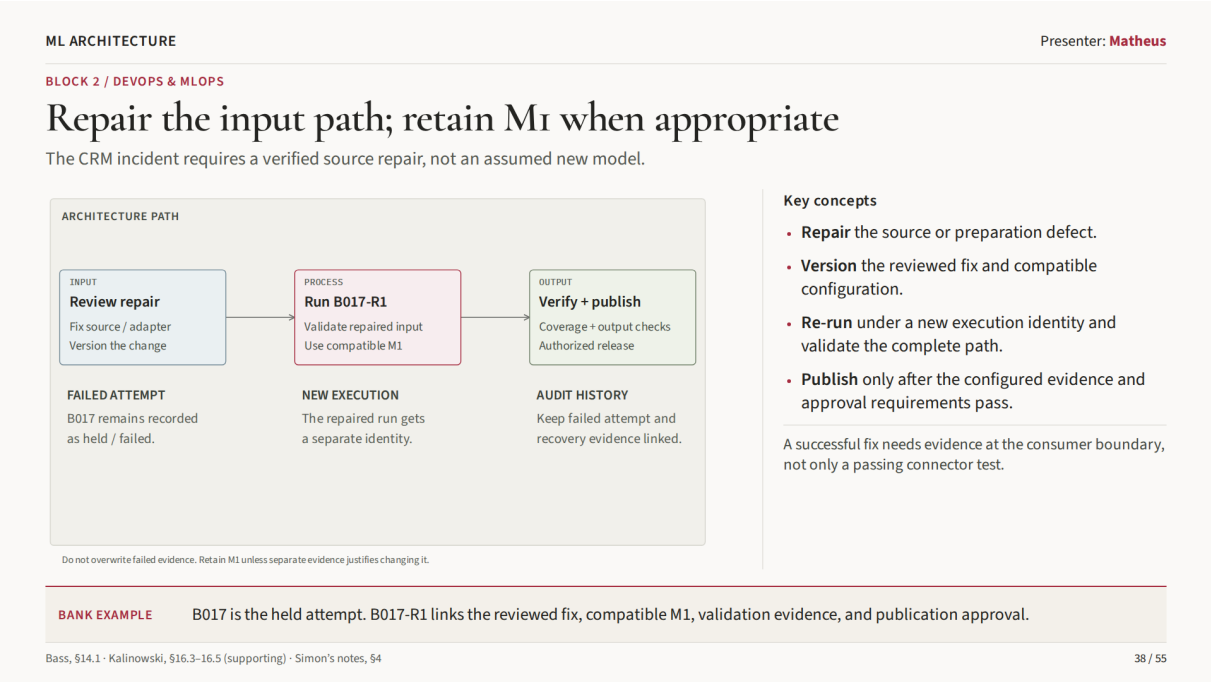
CUES

- Presenter: **Matheus**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

Reading anchor: Bass, §14.1 · Kalinowski, §16.3–16.5 (supporting)

38 / 55 Repair the input path; retain M1 when appropriate

Block 2 / DevOps & MLOps • Matheus



TO SAY

- The CRM incident requires a verified source repair, not an assumed new model.
- **Repair** the source or preparation defect.
- **Version** the reviewed fix and compatible configuration.
- **Re-run** under a new execution identity and validate the complete path.
- **Publish** only after the configured evidence and approval requirements pass.
- *At our bank:* B017 is the held attempt. B017-R1 links the reviewed fix, compatible M1, validation evidence, and publication approval.
- *A successful fix needs evidence at the consumer boundary, not only a passing connector test.*

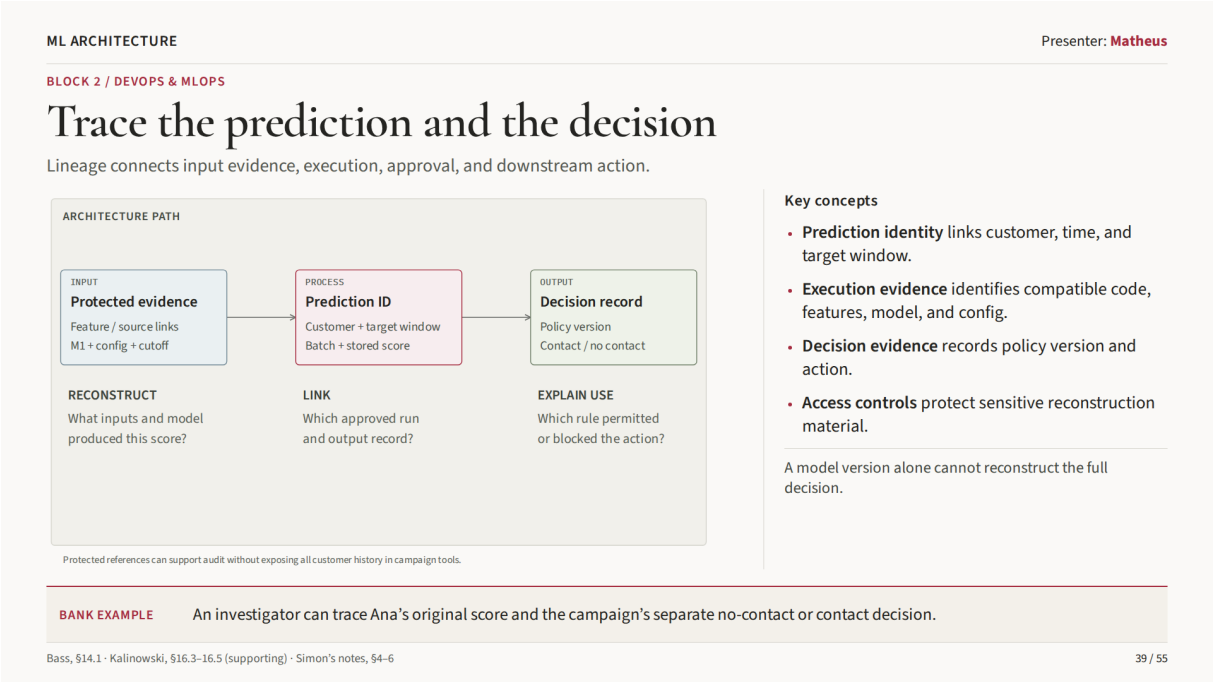
CUES

- Presenter: **Matheus**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

Reading anchor: Bass, §14.1 · Kalinowski, §16.3–16.5 (supporting) · Simon's notes, §4

39 / 55 Trace the prediction and the decision

Block 2 / DevOps & MLOps • Matheus



TO SAY

- Lineage connects input evidence, execution, approval, and downstream action.
- **Prediction identity** links customer, time, and target window.
- **Execution evidence** identifies compatible code, features, model, and config.
- **Decision evidence** records policy version and action.
- **Access controls** protect sensitive reconstruction material.
- *At our bank:* An investigator can trace Ana's original score and the campaign's separate no-contact or contact decision.
- *A model version alone cannot reconstruct the full decision.*

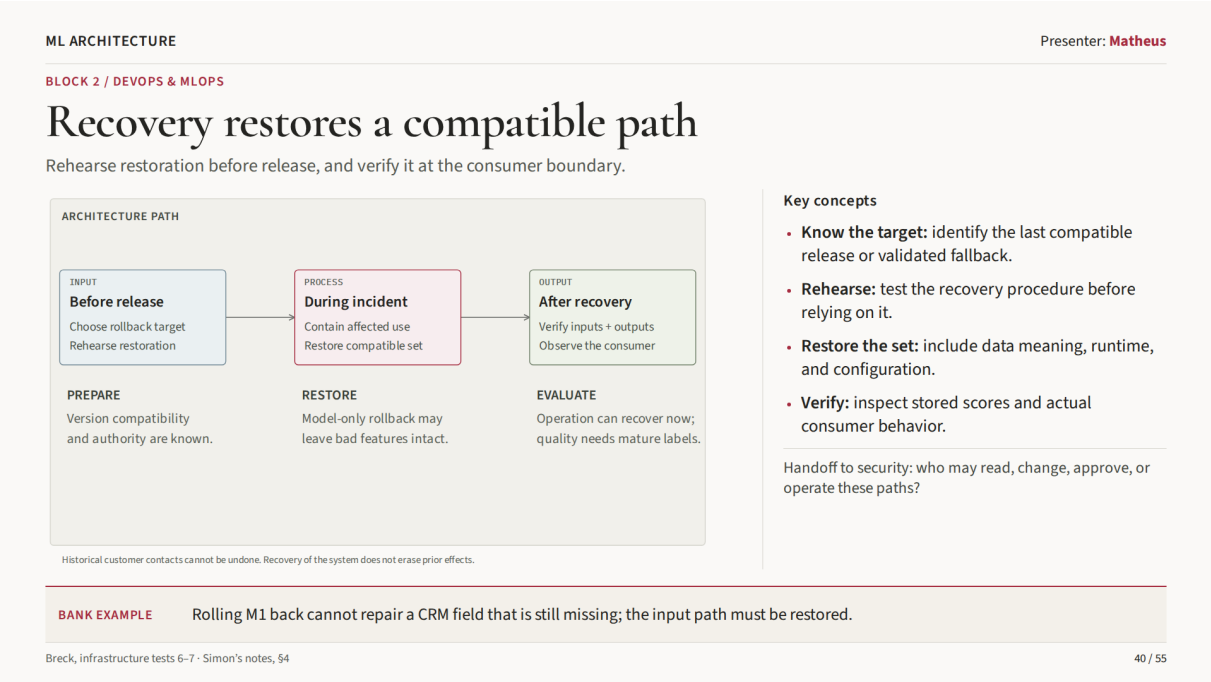
CUES

- Presenter: **Matheus**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

Reading anchor: Bass, §14.1 · Kalinowski, §16.3–16.5 (supporting) · Simon's notes, §4–6

40 / 55 Recovery restores a compatible path

Block 2 / DevOps & MLOps • Matheus



TO SAY

- Rehearse restoration before release, and verify it at the consumer boundary.
- **Know the target:** identify the last compatible release or validated fallback.
- **Rehearse:** test the recovery procedure before relying on it.
- **Restore the set:** include data meaning, runtime, and configuration.
- **Verify:** inspect stored scores and actual consumer behavior.
- *At our bank:* Rolling M1 back cannot repair a CRM field that is still missing; the input path must be restored.
- *Handoff to security: who may read, change, approve, or operate these paths?*

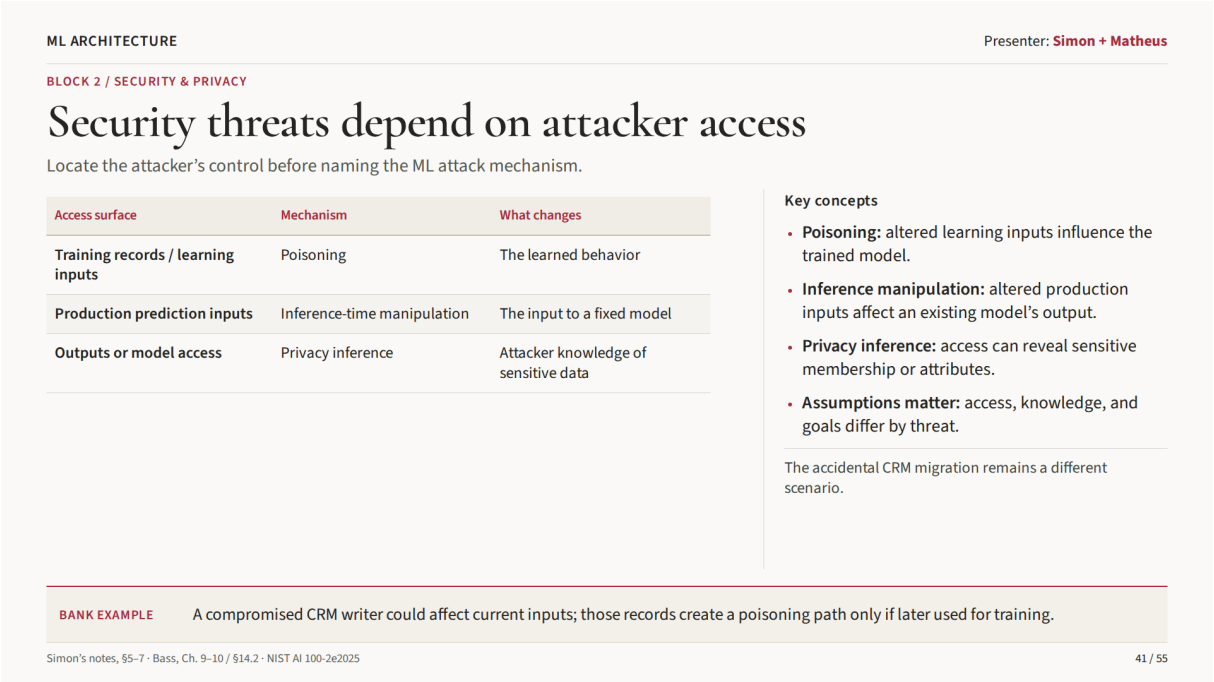
CUES

- Presenter: **Matheus**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

Reading anchor: Breck, infrastructure tests 6–7 · Simon's notes, §4

41 / 55 Security threats depend on attacker access

Block 2 / Security & privacy • Simon + Matheus



TO SAY

- Locate the attacker's control before naming the ML attack mechanism.
- **Poisoning:** altered learning inputs influence the trained model.
- **Inference manipulation:** altered production inputs affect an existing model's output.
- **Privacy inference:** access can reveal sensitive membership or attributes.
- **Assumptions matter:** access, knowledge, and goals differ by threat.
- *At our bank:* A compromised CRM writer could affect current inputs; those records create a poisoning path only if later used for training.
- *The accidental CRM migration remains a different scenario.*

CUES

- New block: **Block 2 / Security & privacy**.
- Presenter: **Simon + Matheus**.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

Reading anchor: Simon's notes, §5–7 · Bass, Ch. 9–10 / §14.2 · NIST AI 100-2e2025

42 / 55 Enforce access at the resource boundary

Block 2 / Security & privacy • Simon + Matheus



TO SAY

- Authentication, authorization, validation, and audit solve different parts of the problem.
- **Authenticate:** establish caller identity.
- **Authorize:** check operation, resource, and scope.
- **Validate:** enforce the input contract independently.
- **Audit:** preserve useful access and change evidence.
- *At our bank:* The scorer may read the approved M1 artifact; it must not gain permission to promote M2 merely by reaching the registry.
- *A trust boundary needs an enforced policy, not only a box on a diagram.*

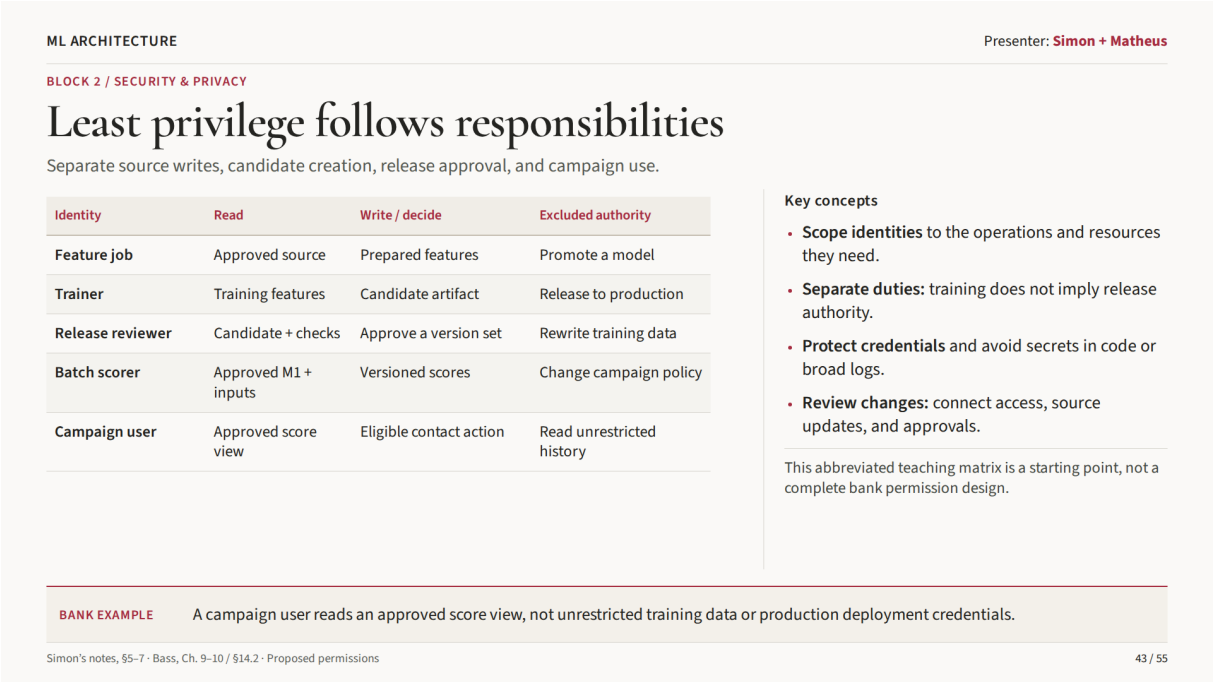
CUES

- Presenter: **Simon + Matheus**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

Reading anchor: Simon's notes, §5–7 · Bass, Ch. 9–10 / §14.2 · NIST SP 800-207, §3

43 / 55 Least privilege follows responsibilities

Block 2 / Security & privacy • Simon + Matheus



TO SAY

- Separate source writes, candidate creation, release approval, and campaign use.
- **Scope identities** to the operations and resources they need.
- **Separate duties:** training does not imply release authority.
- **Protect credentials** and avoid secrets in code or broad logs.
- **Review changes:** connect access, source updates, and approvals.
- *At our bank:* A campaign user reads an approved score view, not unrestricted training data or production deployment credentials.
- *This abbreviated teaching matrix is a starting point, not a complete bank permission design.*

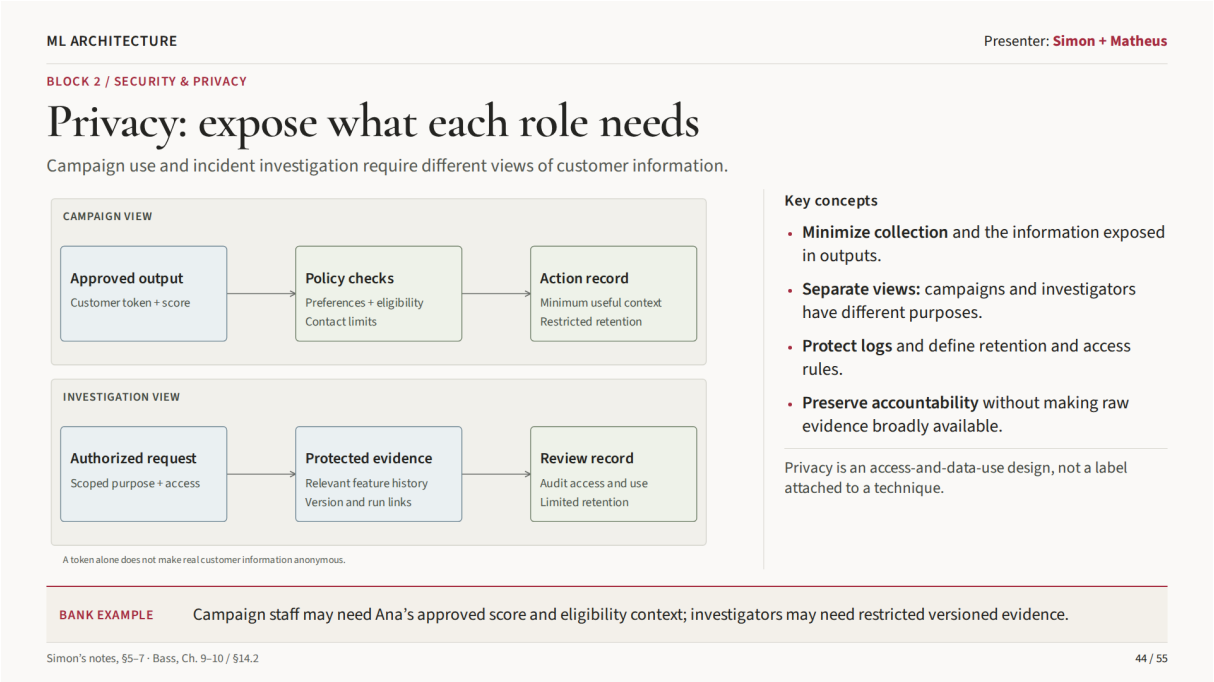
CUES

- Presenter: **Simon + Matheus**.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

Reading anchor: Simon's notes, §5–7 · Bass, Ch. 9–10 / §14.2 · Proposed permissions

44 / 55 Privacy: expose what each role needs

Block 2 / Security & privacy • Simon + Matheus



TO SAY

- Campaign use and incident investigation require different views of customer information.
- **Minimize collection** and the information exposed in outputs.
- **Separate views:** campaigns and investigators have different purposes.
- **Protect logs** and define retention and access rules.
- **Preserve accountability** without making raw evidence broadly available.
- *At our bank:* Campaign staff may need Ana's approved score and eligibility context; investigators may need restricted versioned evidence.
- *Privacy is an access-and-data-use design, not a label attached to a technique.*

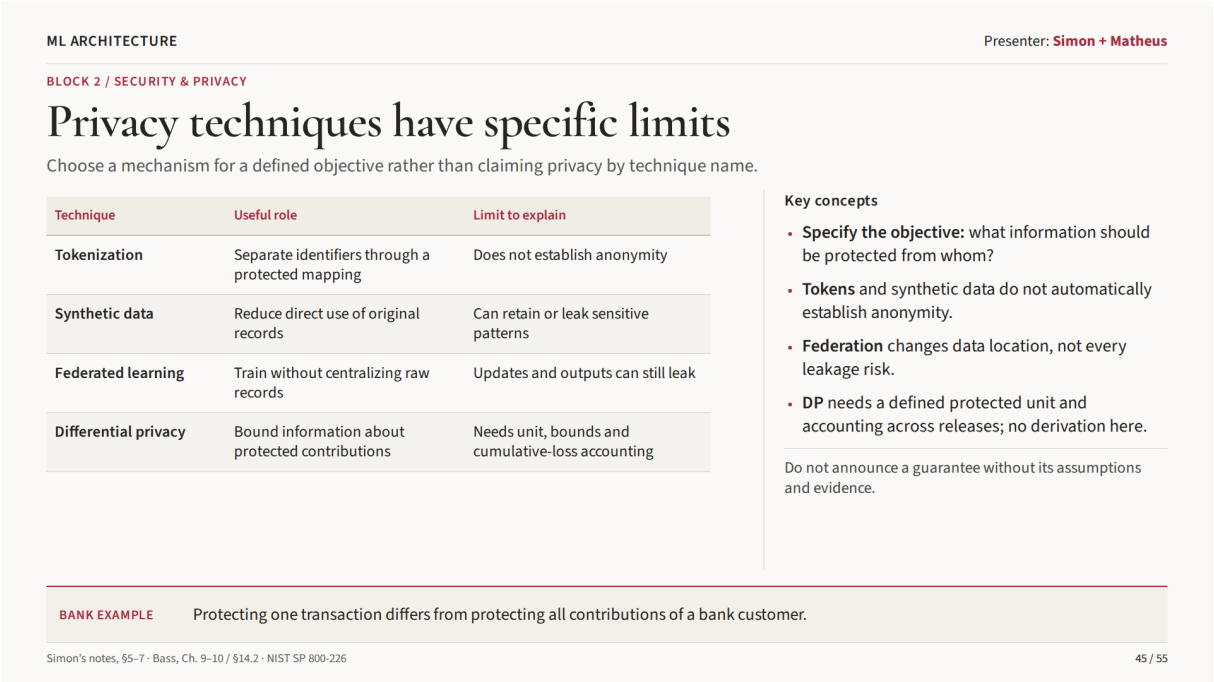
CUES

- Presenter: **Simon + Matheus**.
- Animated build: click 3 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

Reading anchor: Simon's notes, §5–7 · Bass, Ch. 9–10 / §14.2

45 / 55 Privacy techniques have specific limits

Block 2 / Security & privacy • Simon + Matheus



TO SAY

- Choose a mechanism for a defined objective rather than claiming privacy by technique name.
- **Specify the objective:** what information should be protected from whom?
- **Tokens** and synthetic data do not automatically establish anonymity.
- **Federation** changes data location, not every leakage risk.
- **DP** needs a defined protected unit and accounting across releases; no derivation here.
- *At our bank:* Protecting one transaction differs from protecting all contributions of a bank customer.
- *Do not announce a guarantee without its assumptions and evidence.*

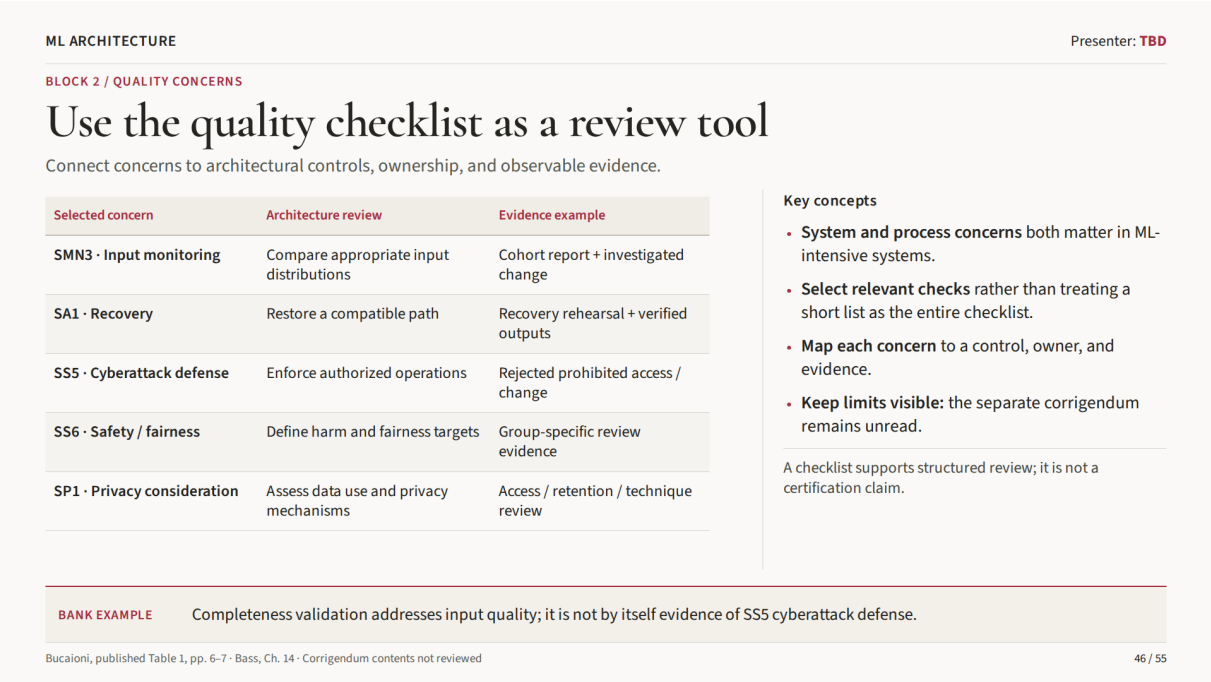
CUES

- Presenter: **Simon + Matheus**.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

Reading anchor: Simon's notes, §5–7 · Bass, Ch. 9–10 / §14.2 · NIST SP 800-226

46 / 55 Use the quality checklist as a review tool

Block 2 / Quality concerns • TBD



TO SAY

- Connect concerns to architectural controls, ownership, and observable evidence.
- **System and process concerns** both matter in ML-intensive systems.
- **Select relevant checks** rather than treating a short list as the entire checklist.
- **Map each concern** to a control, owner, and evidence.
- **Keep limits visible:** the separate corrigendum remains unread.
- *At our bank:* Completeness validation addresses input quality; it is not by itself evidence of SS5 cyberattack defense.
- *A checklist supports structured review; it is not a certification claim.*

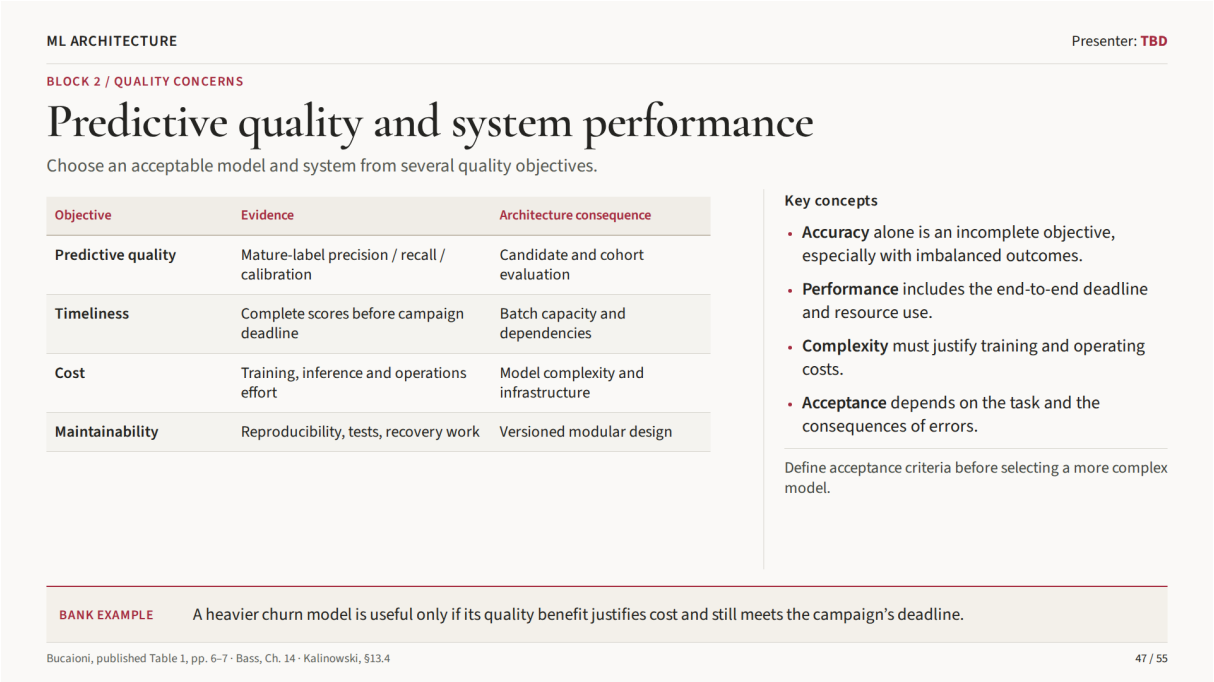
CUES

- New block: **Block 2 / Quality concerns**.
- Presenter: **TBD**.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

Reading anchor: Bucaioni, published Table 1, pp. 6–7 · Bass, Ch. 14 · Corrigendum contents not reviewed

47 / 55 Predictive quality and system performance

Block 2 / Quality concerns • TBD



TO SAY

- Choose an acceptable model and system from several quality objectives.
- **Accuracy** alone is an incomplete objective, especially with imbalanced outcomes.
- **Performance** includes the end-to-end deadline and resource use.
- **Complexity** must justify training and operating costs.
- **Acceptance** depends on the task and the consequences of errors.
- *At our bank:* A heavier churn model is useful only if its quality benefit justifies cost and still meets the campaign's deadline.
- *Define acceptance criteria before selecting a more complex model.*

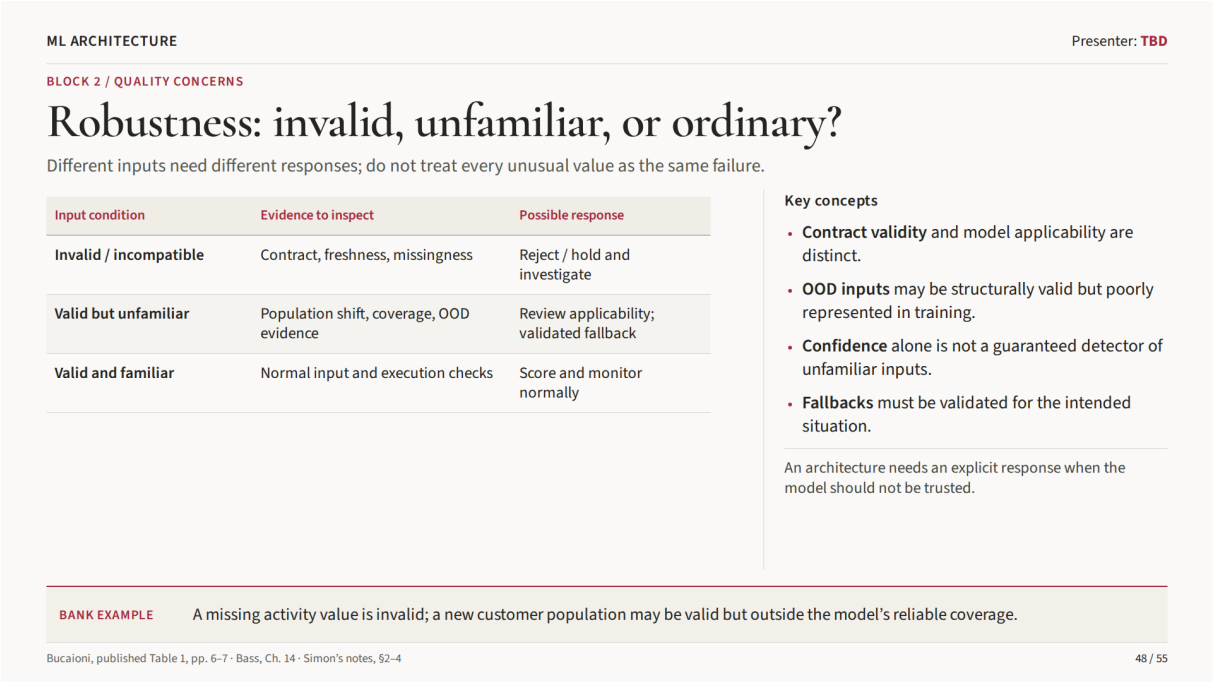
CUES

- Presenter: **TBD**.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

Reading anchor: Bucaioni, published Table 1, pp. 6–7 · Bass, Ch. 14 · Kalinowski, §13.4

48 / 55 Robustness: invalid, unfamiliar, or ordinary?

Block 2 / Quality concerns • TBD



TO SAY

- Different inputs need different responses; do not treat every unusual value as the same failure.
- **Contract validity** and model applicability are distinct.
- **OOD inputs** may be structurally valid but poorly represented in training.
- **Confidence** alone is not a guaranteed detector of unfamiliar inputs.
- **Fallbacks** must be validated for the intended situation.
- *At our bank:* A missing activity value is invalid; a new customer population may be valid but outside the model's reliable coverage.
- *An architecture needs an explicit response when the model should not be trusted.*

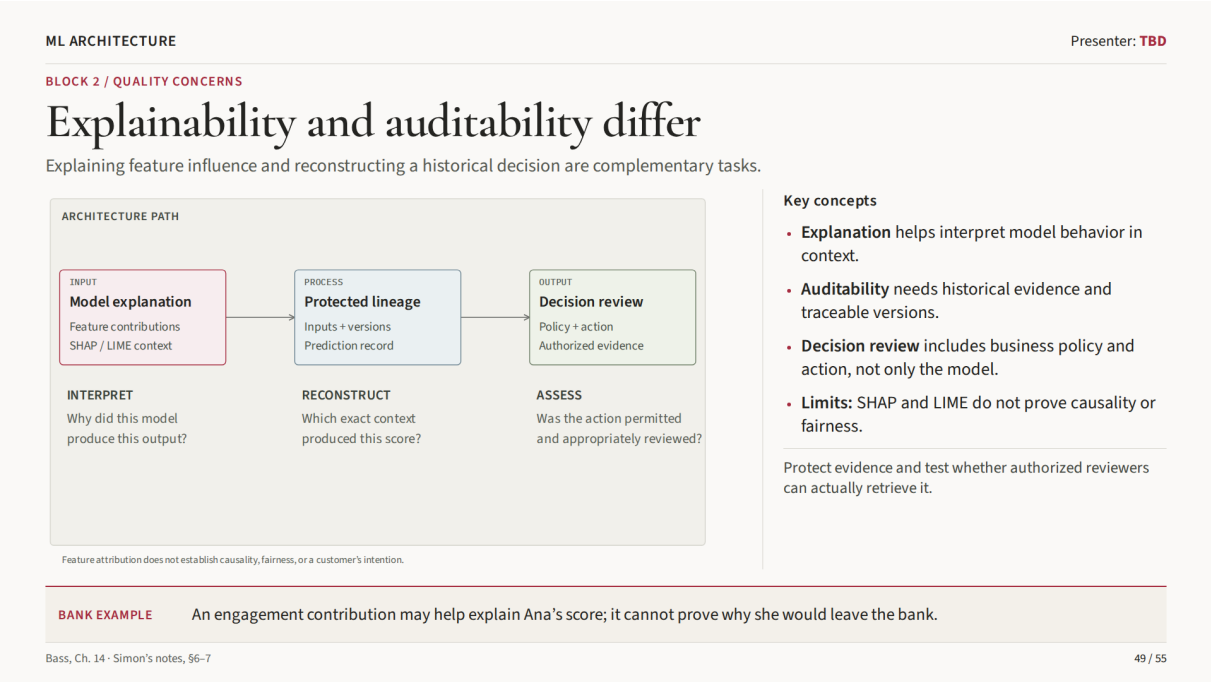
CUES

- Presenter: **TBD**.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

Reading anchor: Bucaioni, published Table 1, pp. 6–7 · Bass, Ch. 14 · Simon's notes, §2–4

49 / 55 Explainability and auditability differ

Block 2 / Quality concerns • TBD



TO SAY

- Explaining feature influence and reconstructing a historical decision are complementary tasks.
- **Explanation** helps interpret model behavior in context.
- **Auditability** needs historical evidence and traceable versions.
- **Decision review** includes business policy and action, not only the model.
- **Limits:** SHAP and LIME do not prove causality or fairness.
- *At our bank:* An engagement contribution may help explain Ana's score; it cannot prove why she would leave the bank.
- *Protect evidence and test whether authorized reviewers can actually retrieve it.*

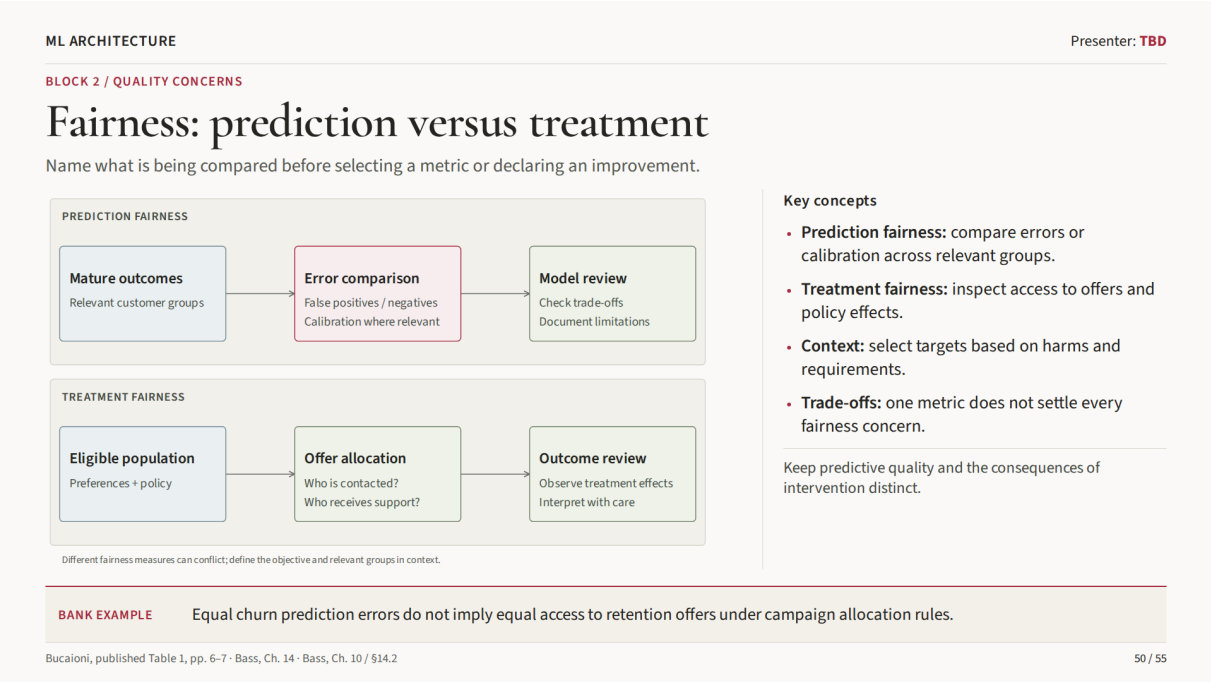
CUES

- Presenter: **TBD**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

Reading anchor: Bass, Ch. 14 · Simon's notes, §6–7

50 / 55 Fairness: prediction versus treatment

Block 2 / Quality concerns • TBD



TO SAY

- Name what is being compared before selecting a metric or declaring an improvement.
- **Prediction fairness:** compare errors or calibration across relevant groups.
- **Treatment fairness:** inspect access to offers and policy effects.
- **Context:** select targets based on harms and requirements.
- **Trade-offs:** one metric does not settle every fairness concern.
- *At our bank:* Equal churn prediction errors do not imply equal access to retention offers under campaign allocation rules.
- *Keep predictive quality and the consequences of intervention distinct.*

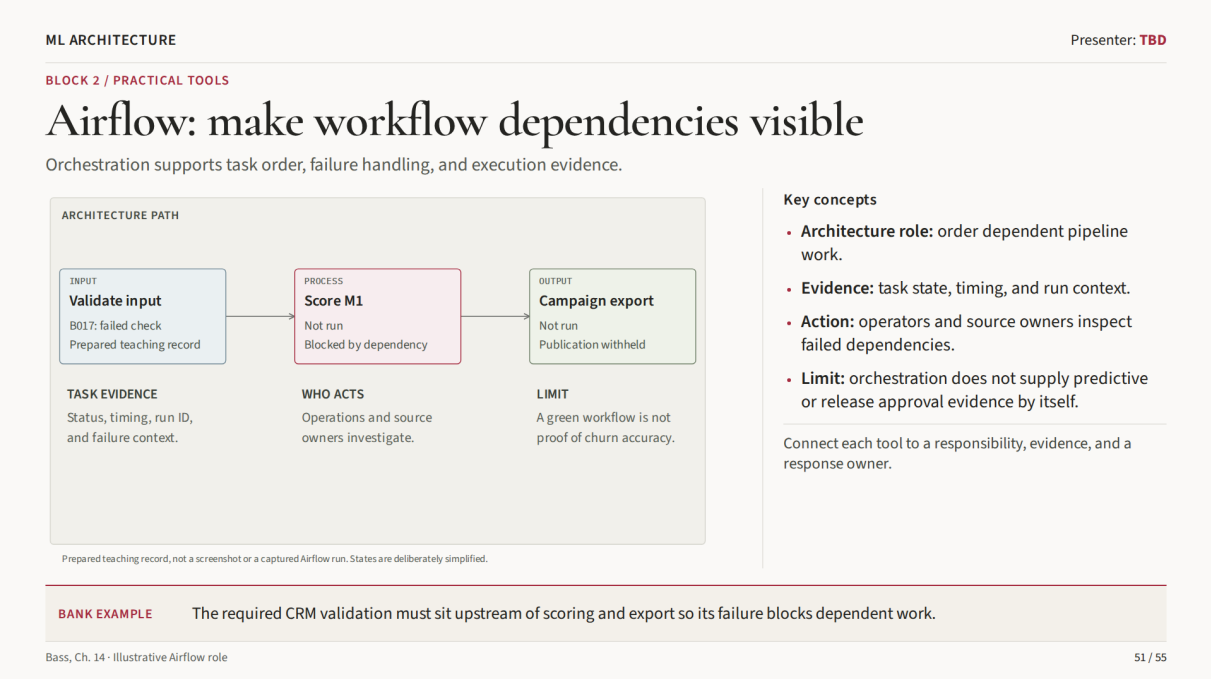
CUES

- Presenter: **TBD**.
- Animated build: click 3 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

Reading anchor: Bucaioni, published Table 1, pp. 6–7 · Bass, Ch. 14 · Bass, Ch. 10 / §14.2

51 / 55 Airflow: make workflow dependencies visible

Block 2 / Practical tools • TBD



TO SAY

- Orchestration supports task order, failure handling, and execution evidence.
- **Architecture role:** order dependent pipeline work.
- **Evidence:** task state, timing, and run context.
- **Action:** operators and source owners inspect failed dependencies.
- **Limit:** orchestration does not supply predictive or release approval evidence by itself.
- *At our bank:* The required CRM validation must sit upstream of scoring and export so its failure blocks dependent work.
- *Connect each tool to a responsibility, evidence, and a response owner.*

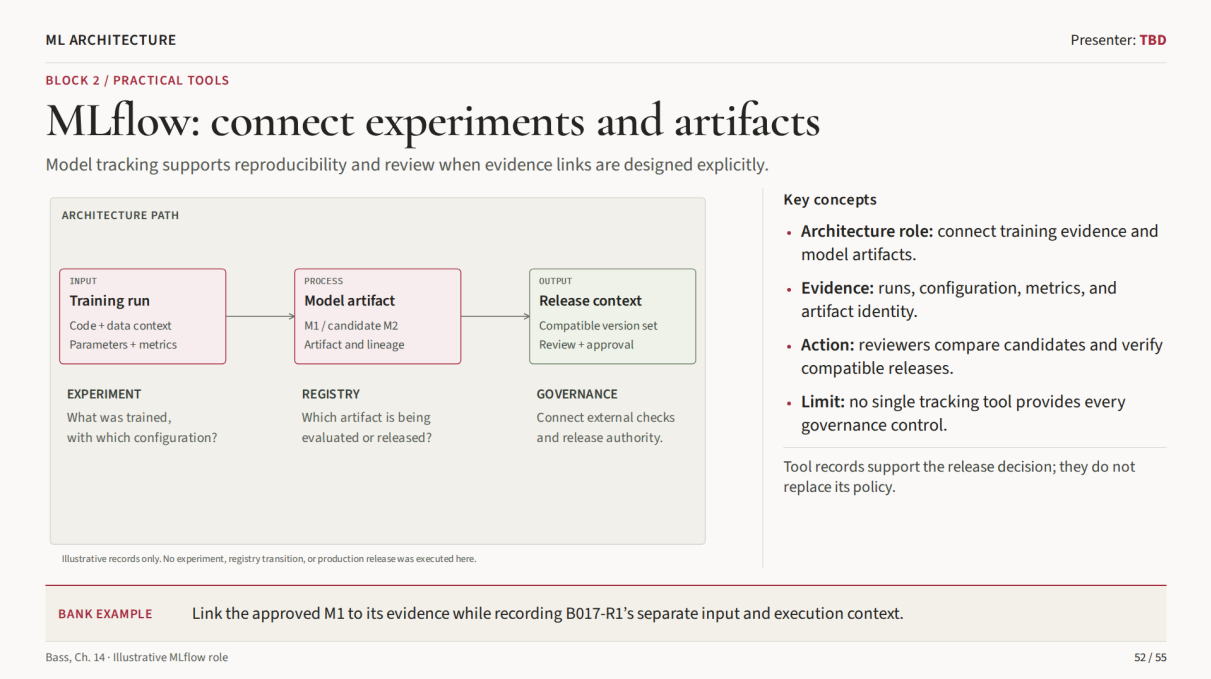
CUES

- New block: **Block 2 / Practical tools**.
- Presenter: **TBD**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

Reading anchor: Bass, Ch. 14 · Illustrative Airflow role

52 / 55 MLflow: connect experiments and artifacts

Block 2 / Practical tools • TBD



TO SAY

- Model tracking supports reproducibility and review when evidence links are designed explicitly.
- **Architecture role:** connect training evidence and model artifacts.
- **Evidence:** runs, configuration, metrics, and artifact identity.
- **Action:** reviewers compare candidates and verify compatible releases.
- **Limit:** no single tracking tool provides every governance control.
- *At our bank:* Link the approved M1 to its evidence while recording B017-R1's separate input and execution context.
- *Tool records support the release decision; they do not replace its policy.*

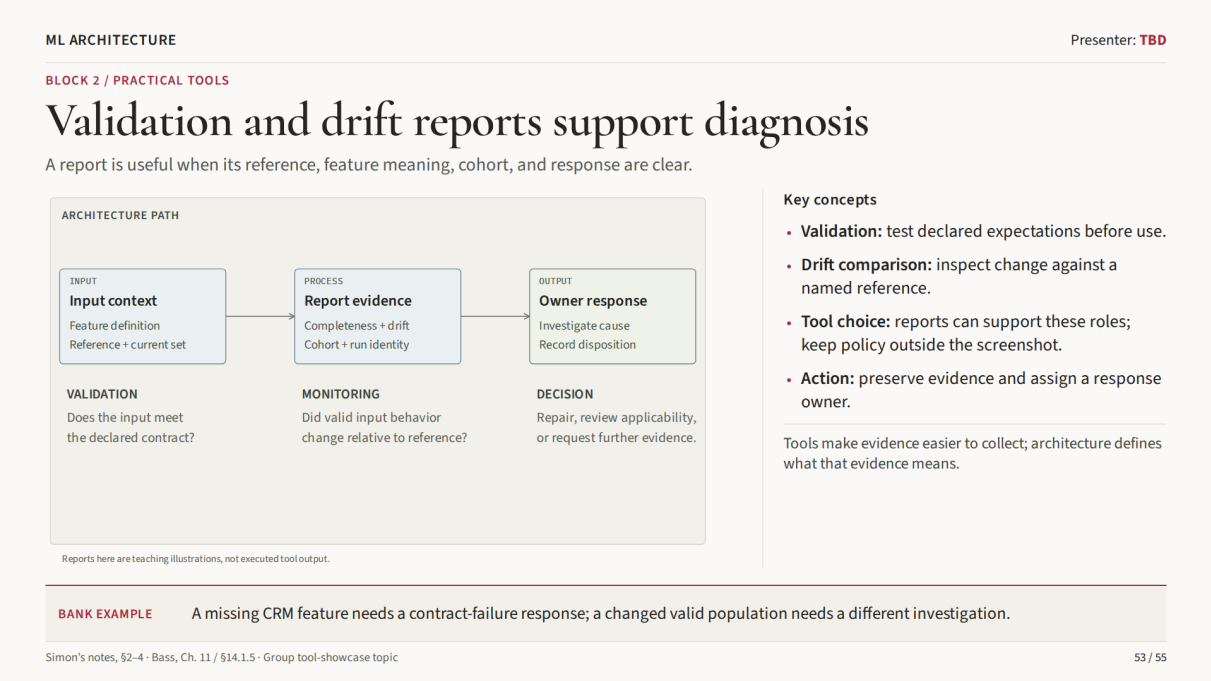
CUES

- Presenter: **TBD**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

Reading anchor: Bass, Ch. 14 · Illustrative MLflow role

53 / 55 Validation and drift reports support diagnosis

Block 2 / Practical tools • TBD



TO SAY

- A report is useful when its reference, feature meaning, cohort, and response are clear.
- **Validation:** test declared expectations before use.
- **Drift comparison:** inspect change against a named reference.
- **Tool choice:** reports can support these roles; keep policy outside the screenshot.
- **Action:** preserve evidence and assign a response owner.
- *At our bank:* A missing CRM feature needs a contract-failure response; a changed valid population needs a different investigation.
- *Tools make evidence easier to collect; architecture defines what that evidence means.*

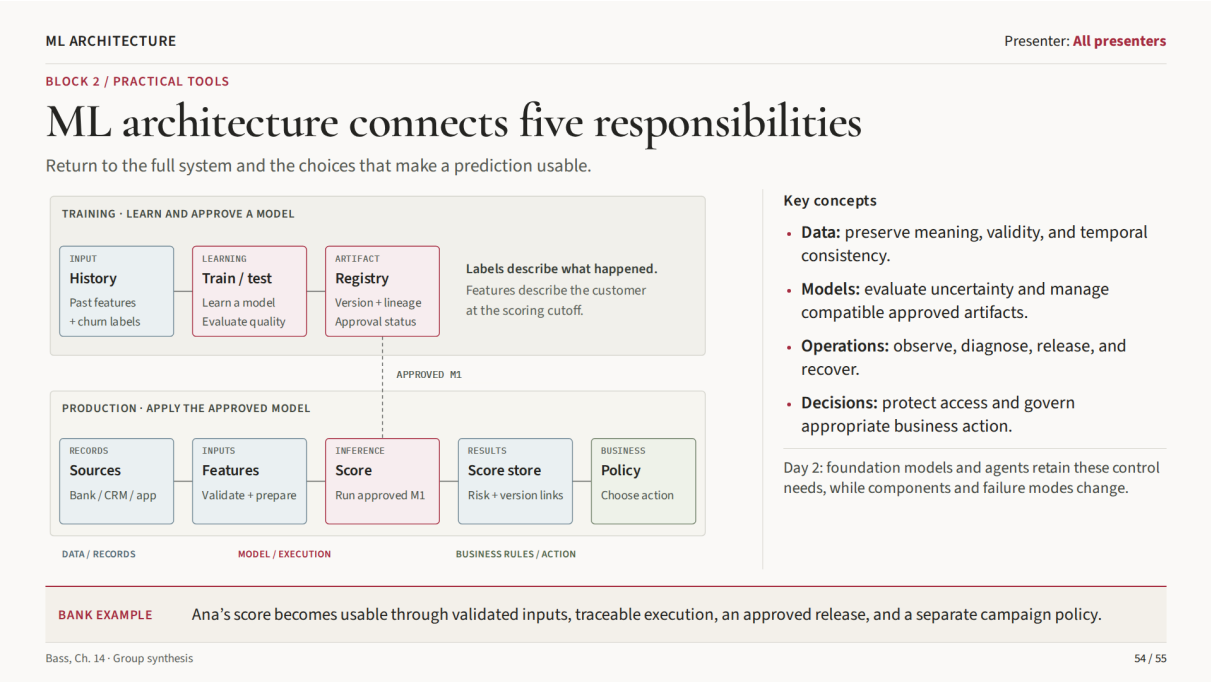
CUES

- Presenter: **TBD**.
- Animated build: click 2 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

Reading anchor: Simon's notes, §2–4 · Bass, Ch. 11 / §14.1.5 · Group tool-showcase topic

54 / 55 ML architecture connects five responsibilities

Block 2 / Practical tools • All presenters



TO SAY

- Return to the full system and the choices that make a prediction usable.
- **Data:** preserve meaning, validity, and temporal consistency.
- **Models:** evaluate uncertainty and manage compatible approved artifacts.
- **Operations:** observe, diagnose, release, and recover.
- **Decisions:** protect access and govern appropriate business action.
- *At our bank:* Ana's score becomes usable through validated inputs, traceable execution, an approved release, and a separate campaign policy.
- *Day 2: foundation models and agents retain these control needs, while components and failure modes change.*

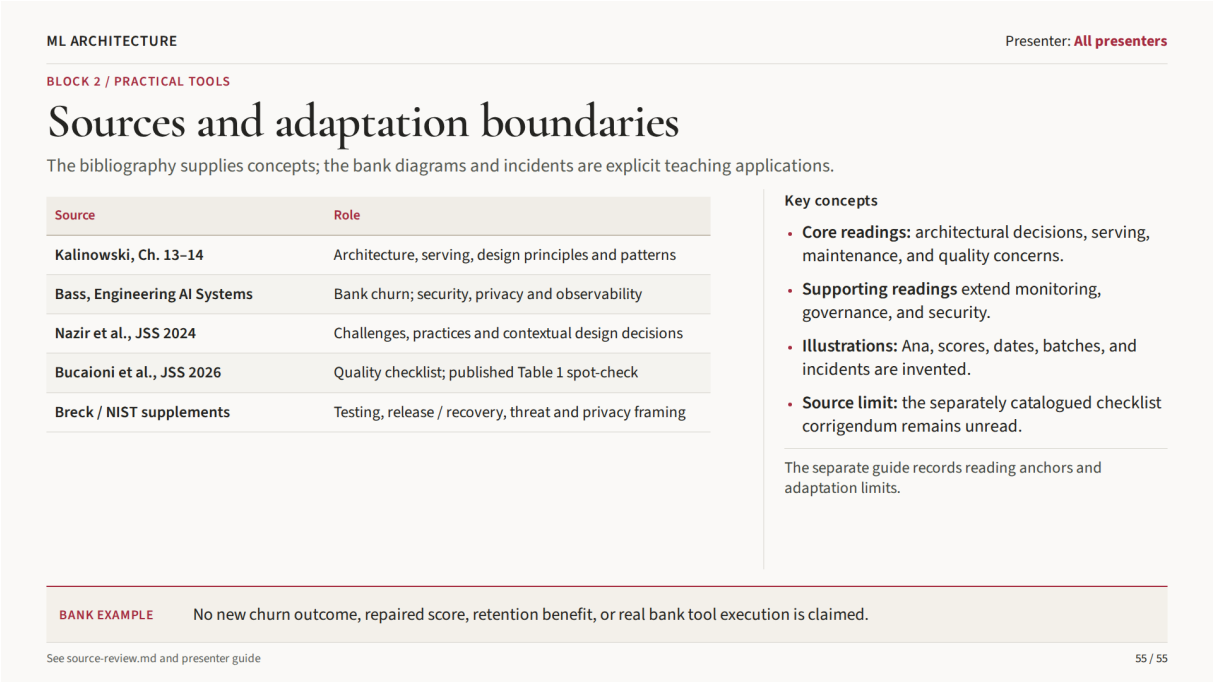
CUES

- Presenter: **All presenters**.
- Animated build: click 4 times to reveal the remaining steps before moving on.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

Reading anchor: Bass, Ch. 14 · Group synthesis

55 / 55 Sources and adaptation boundaries

Block 2 / Practical tools • All presenters



TO SAY

- The bibliography supplies concepts; the bank diagrams and incidents are explicit teaching applications.
- **Core readings:** architectural decisions, serving, maintenance, and quality concerns.
- **Supporting readings** extend monitoring, governance, and security.
- **Illustrations:** Ana, scores, dates, batches, and incidents are invented.
- **Source limit:** the separately catalogued checklist corrigendum remains unread.
- *At our bank:* No new churn outcome, repaired score, retention benefit, or real bank tool execution is claimed.
- *The separate guide records reading anchors and adaptation limits.*

CUES

- Presenter: **All presenters**.
- Pause briefly before the bank example — let the concept land first.
- Close on the takeaway line before advancing.

Reading anchor: See source-review.md and presenter guide